HRMS

03 考勤管理 — 系統設計書

HR03 ATTENDANCE Service — System Design

系統名稱 **人力資源暨專案管理系統**	版本 **1.0**
日期 **2026-03-17**	作者 **HRMS 開發團隊**

目錄

考勤管理服務系統設計書

版本: 1.0 **日期:** 2025-12-06 **Domain代號:** 03 (ATT) **目標:** 提供工程師完整的系統實作規格，供PM建立工項清單

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1. 服務概述

1.1 服務定位

考勤管理服務負責員工的出勤管理、假勤申請與審核，以及加班管理。本服務必須確保**符合台灣勞動基準法**的所有規定。

1.2 核心功能

- ✅ **打卡管理:** 多元打卡方式、打卡異常偵測、補卡申請
- ✅ **班別設定:** 彈性班別、輪班、彈性工時管理
- ✅ **假勤管理:** 20+種假別設定、請假申請與審核、假期餘額計算
- ✅ **特休假管理:** 依勞基法自動計算特休、到期提醒、未休補償
- ✅ **加班管理:** 加班申請審核、加班時數管控（46小時/月、138小時/3月）
- ✅ **變形工時:** 支援二週/四週/八週變形工時排班
- ✅ **法規遵循:** 女性員工保護、職業災害管理

- ✅ **差勤報表:** 個人/部門月度差勤統計

1.3 技術架構

- **前端:** ReactJS + Redux + Ant Design
- **後端:** Spring Boot 3.1.x + MyBatis
- **資料庫:** PostgreSQL 15.x
- **事件匯流排:** Kafka
- **排程任務:** Quartz Scheduler

1.4 服務邊界

屬於本服務	不屬於本服務
打卡記錄	薪資計算 (Payroll Service使用本服務數據)
請假申請與審核	員工基本資料 (Organization Service)
加班申請與審核	審核流程引擎 (Workflow Service)
假期餘額計算	
班別與假別規則設定	

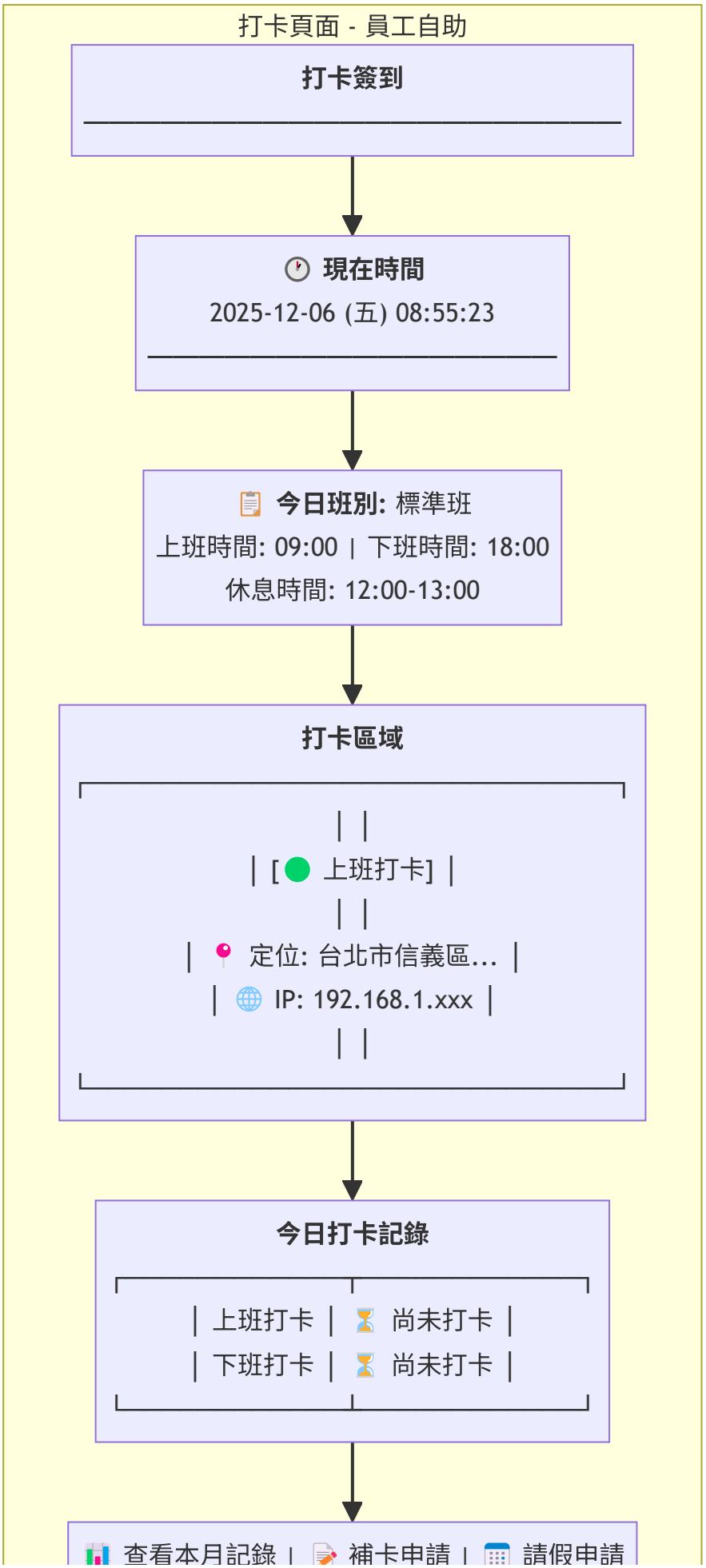
2. UI設計

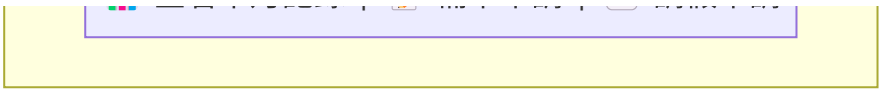
2.1 頁面清單

頁面代碼	頁面名稱	路由	權限要求
HR03-P01	打卡頁面	/attendance/check-in	-
HR03-P02	我的差勤記錄頁面	/attendance/my-records	-
HR03-P03	請假申請頁面	/attendance/leave/apply	-
HR03-P04	我的假期餘額頁面	/attendance/leave/balance	-
HR03-P05	加班申請頁面	/attendance/overtime/apply	-
HR03-P06	差勤審核頁面 (主管)	/admin/attendance/approvals	attendance:approve
HR03-P07	班別管理頁面	/admin/attendance/shifts	shift:manage
HR03-P08	假別管理頁面	/admin/attendance/leave-types	leave-type:manage
HR03-P09	部門差勤報表頁面	/admin/attendance/reports	attendance:report
HR03-P10	月度差勤結算頁面	/admin/attendance/monthly-close	attendance:close

2.2 UI線稿 (Mermaid)

2.2.1 打卡頁面 (HR03-P01)


打卡頁面 - 員工自助
打卡簽到
🕐 現在時間
2025-12-06 (五) 08:55:23
📋 今日班別: 標準班
上班時間: 09:00 | 下班時間: 18:00
休息時間: 12:00-13:00
打卡區域
[🟢 上班打卡]
📍 定位: 台北市信義區...
🌐 IP: 192.168.1.xxx
今日打卡記錄
上班打卡 ⏳ 尚未打卡
下班打卡 ⏳ 尚未打卡
📊 查看本月記錄 | 📝 補卡申請 | 📅 請假申請



頁面元素說明:

- **時間顯示:** 即時更新的系統時間
- **班別資訊:** 顯示員工今日排定的班別
- **打卡按鈕:**
  - 上班時段顯示「上班打卡」(綠色)
  - 下班時段顯示「下班打卡」(藍色)
  - 已打卡顯示「已打卡 ✓」(灰色)
- **定位資訊:** 顯示GPS定位與IP
- **今日記錄:** 即時顯示上/下班打卡狀態

元件規格:

```
interface CheckInPageData {
  currentTime: Date;
  shift: {
    shiftName: string;
    workStartTime: string;
    workEndTime: string;
    breakStartTime: string;
    breakEndTime: string;
  };
  todayRecord: {
    checkInTime: string | null;
    checkOutTime: string | null;
    isLate: boolean;
    lateMinutes: number;
  };
  location: {
    latitude: number;
    longitude: number;
    address: string;
  };
  ipAddress: string;
}
```

2.2.2 我的差勤記錄頁面 (HR03-P02)

我的差勤記錄頁面

我的差勤記錄

📅 月份: [2025年12月▼] | 🔄 重新整理

本月統計

工作日	實際出勤	請假天數	加班時數
22天	18天	2天	8小時

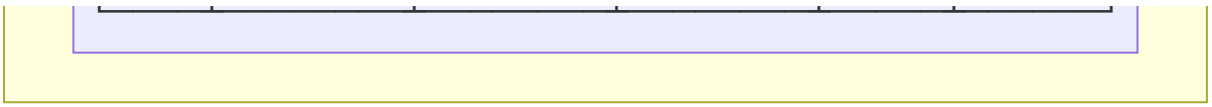
差勤行事曆

日	一	二	三	四	五	六
1	2	3	4	5	6	
✓	✓	✓	✓	✓	—	
7	8	9	10	11	12	13
—	✓	✓	🔴	✓	✓	—
			(遲到5分)			

圖例: ✓正常 🔴異常 📝請假 💼加班 —休息

詳細記錄

日期	上班打卡	下班打卡	工時	狀態	操作
12/6	09:00	18:05	8.08h	✓正常	
12/5	09:03	18:00	7.95h	✓正常	
12/4	09:10	18:00	7.83h	🔴遲到	[補卡]



2.2.3 請假申請頁面 (HR03-P03)

請假申請頁面

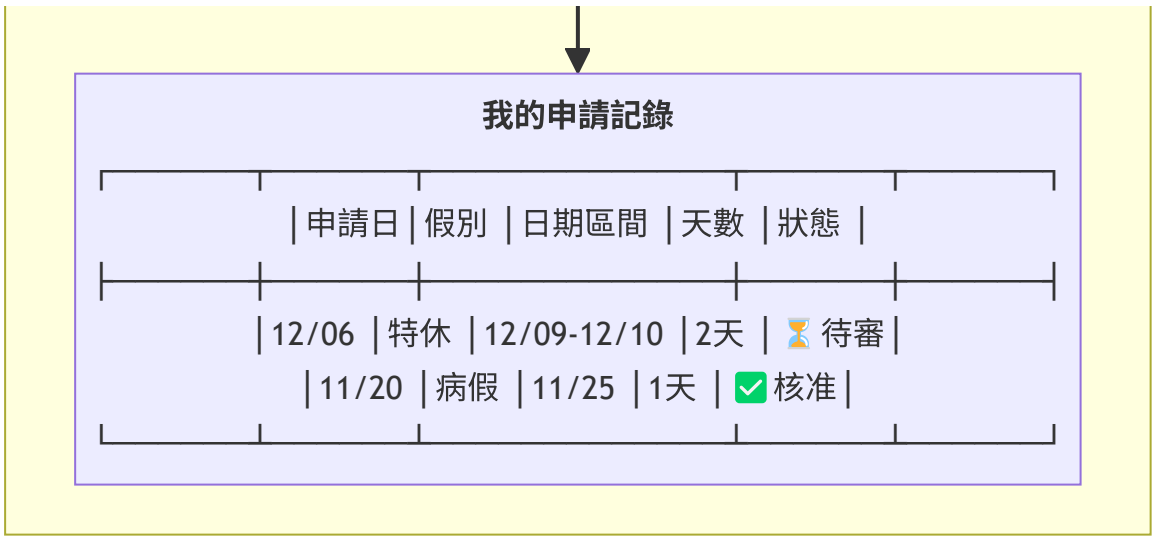
請假申請

假期餘額

假別	剩餘	年度額
🏖️ 特休假	6.5天	10天
🤒 病假	28天	30天
📝 事假	14天	14天
💒 婚假	8天	8天

請假表單

```
┌──────────────────────────────────────┐
│ * 假別類型: [特休假▼] │
│ │
│ * 請假日期: │
│ 起始: [2025-12-09📅] [全天▼] │
│ 結束: [2025-12-10📅] [全天▼] │
│ │
│ 預計請假: 2 天 │
│ 扣除後餘額: 4.5 天 │
│ │
│ * 請假事由: │
│ [____________________] │
│ │
│ 附件上傳: [選擇檔案...] (選填)│
│ │
│ [取消] [儲存草稿] [送出申請] │
└──────────────────────────────────────┘
```

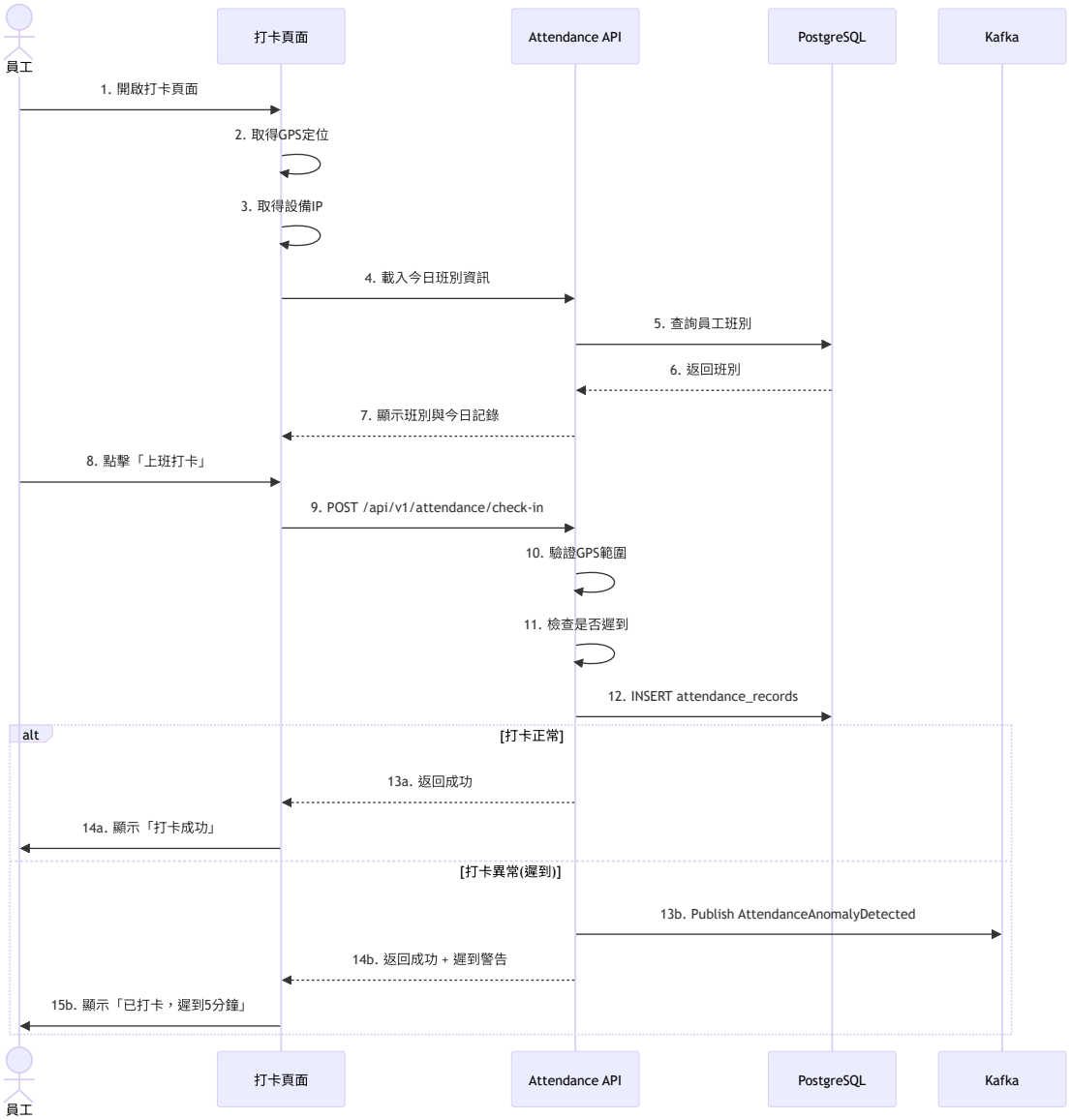

我的申請記錄
申請日
假別
日期區間
天數
狀態
12/06
特休
12/09-12/10
2天
待審
11/20
病假
11/25
1天
核准

2.2.4 差勤審核頁面 (HR03-P06)

```
┌─────────────────────────────────────────────────────────────────────────────┐
│                              差勤審核頁面 - 主管專用                          │
│                                                                             │
│                              **差勤審核**                                    │
│                              ═══════════                                    │
│                                                                             │
│                                    ↓                                        │
│                                                                             │
│                    📝 請假審核 (3) | 💼 加班審核 (1) | 🔧 補卡審核 (2)        │
│                                                                             │
│                                    ↓                                        │
│                                                                             │
│                          **待審核請假申請**                                  │
│                                                                             │
│  ┌──────┬──────────┬──────┬──────────────┬──────┬──────────────┐           │
│  │申請人│申請日期  │假別  │日期區間      │天數  │操作          │           │
│  ├──────┼──────────┼──────┼──────────────┼──────┼──────────────┤           │
│  │張三  │12/06     │特休  │12/09-12/10   │2天   │[核准][駁回]  │           │
│  │李四  │12/05     │病假  │12/08         │1天   │[核准][駁回]  │           │
│  │王五  │12/04     │事假  │12/07         │0.5天 │[核准][駁回]  │           │
│  └──────┴──────────┴──────┴──────────────┴──────┴──────────────┘           │
│                                                                             │
│                                    ↓                                        │
│                                                                             │
│                  **申請詳情** (點擊申請後顯示)                               │
│                                                                             │
│                  ┌──────────────────────────────────┐                       │
│                  │ 申請人: 張三                     │                       │
│                  │ 假別: 特休假                     │                       │
│                  │ 日期: 2025/12/09 - 2025/12/10    │                       │
│                  │ 天數: 2天                        │                       │
│                  │ 事由: 家庭事務                   │                       │
│                  │ 餘額: 6.5天 → 4.5天              │                       │
│                  │                                  │                       │
│                  │ 駁回原因: [_____________]        │                       │
│                  │                                  │                       │
│                  │ [核准申請] [駁回申請]            │                       │
│                  └──────────────────────────────────┘                       │
└─────────────────────────────────────────────────────────────────────────────┘
```

3. UX流程設計

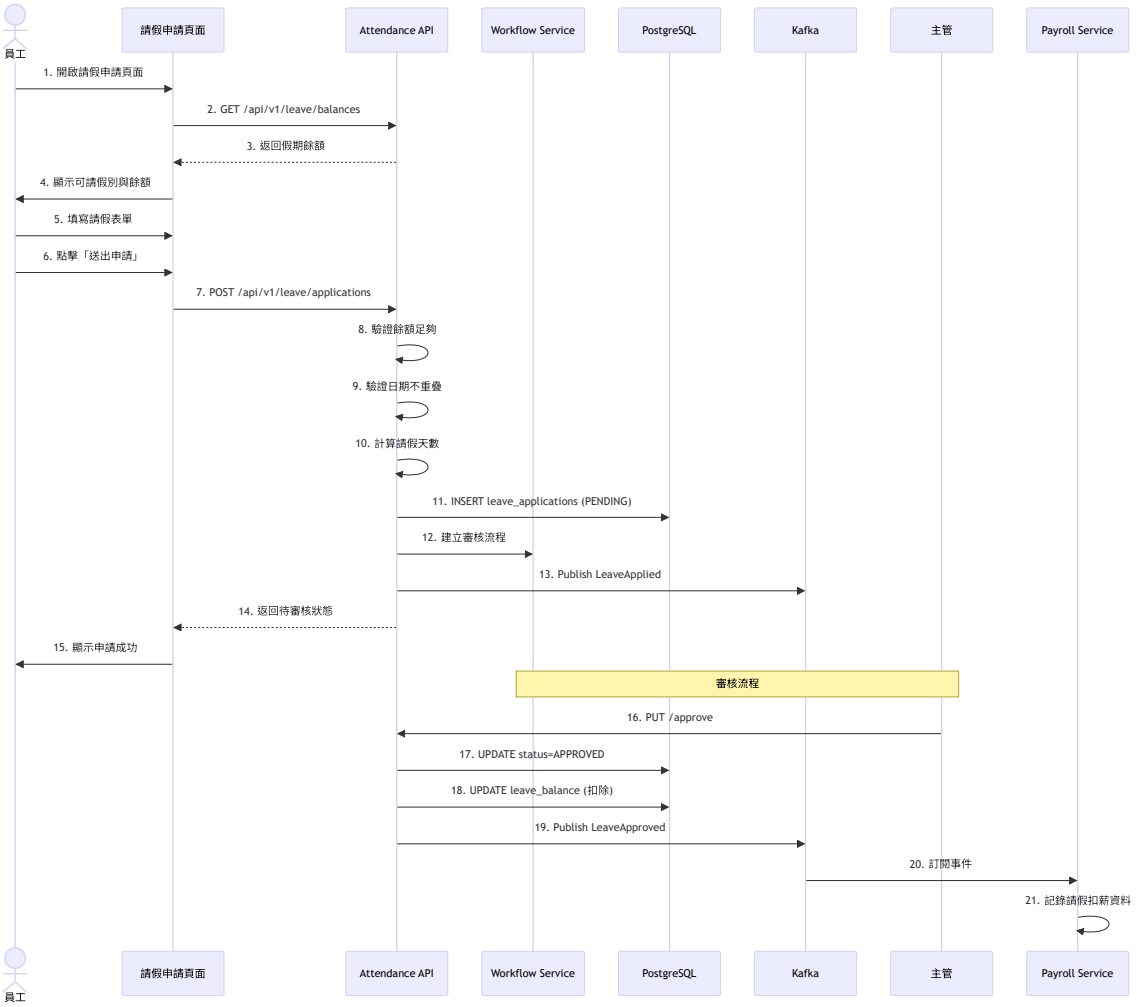
3.1 員工打卡流程



關鍵點:

- ✅ GPS定位驗證 (誤差<50m)
- ✅ 自動偵測遲到/早退
- ✅ 異常打卡發布事件通知

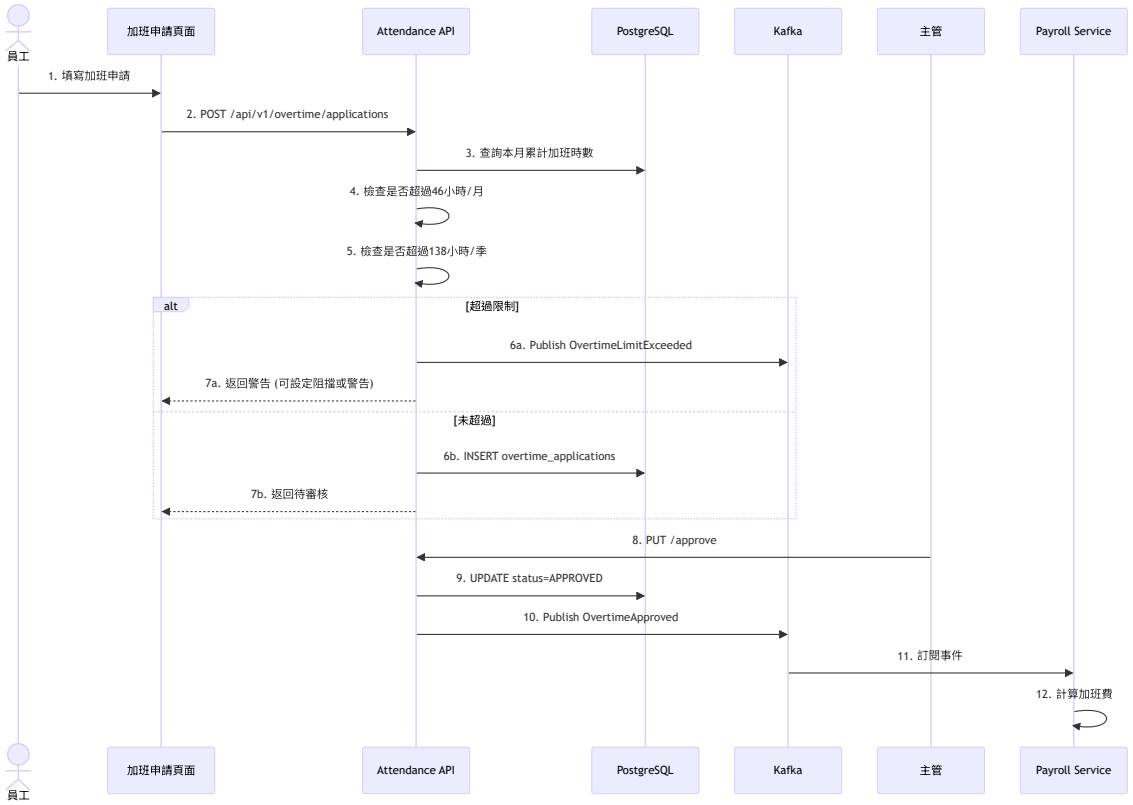
3.2 請假申請流程



關鍵點:

- ✅ 即時餘額驗證
- ✅ 重疊日期檢查
- ✅ 整合Workflow審核流程
- ✅ 核准後自動扣除餘額
- ✅ 發布事件供Payroll計算

3.3 加班申請流程



關鍵點:

- ✅ 勞基法加班時數管控 (46h/月、138h/季)
- ✅ 超時申請預警機制
- ✅ 審核後自動計算加班費/補休

(文件持續，下一部分包含畫面事件說明、資料庫設計、Domain設計、API規格等)

Continued: 03_考勤管理服務系統設計書_part2.md

4. 畫面事件說明

4.1 打卡頁面事件 (HR03-P01)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-ATT-01`	頁面載入	onMount	取得GPS定位 + 載入班別	GET /api/v1/attendance/today
`E-ATT-02`	上班打卡按鈕	onClick	執行上班打卡	POST /api/v1/attendance/check-in
`E-ATT-03`	下班打卡按鈕	onClick	執行下班打卡	POST /api/v1/attendance/check-out
`E-ATT-04`	查看本月記錄	onClick	路由跳轉	-
`E-ATT-05`	補卡申請	onClick	開啟補卡對話框	-

E-ATT-02 詳細流程:

```
const handleCheckIn = async () => {
  try {
    // 1. 取得當前位置
    const position = await getCurrentPosition();

    // 2. 組裝請求資料
    const request = {
      employeeId: currentUser.employeeId,
      checkInTime: new Date().toISOString(),
      location: {
        latitude: position.coords.latitude,
        longitude: position.coords.longitude
      },
      ipAddress: await getClientIP()
    };

    // 3. 呼叫API
    setLoading(true);
    const response = await attendanceService.checkIn(request);

    // 4. 更新本地狀態
    dispatch(setTodayRecord(response));

    // 5. 顯示結果
    if (response.isLate) {
      message.warning(`打卡成功，遲到 ${response.lateMinutes} 分鐘`);
    } else {
      message.success('打卡成功！');
    }

  } catch (error) {
    if (error.code === 'LOCATION_OUT_OF_RANGE') {
      message.error('您的位置不在允許的打卡範圍內');
    } else {
      message.error('打卡失敗，請稍後重試');
    }
  } finally {
    setLoading(false);
  }
};
```

4.2 差勤記錄頁面事件 (HR03-P02)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-REC-01`	月份選擇器	onChange	重新載入該月記錄	GET /api/v1/attendance/records?month={month}
`E-REC-02`	日曆日期點擊	onClick	顯示該日詳情	- (本地篩選)
`E-REC-03`	補卡按鈕	onClick	開啟補卡申請對話框	-
`E-REC-04`	補卡送出	onClick	提交補卡申請	POST /api/v1/attendance/corrections

4.3 請假申請頁面事件 (HR03-P03)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-LV-01`	頁面載入	onMount	載入假期餘額	GET /api/v1/leave/balances
`E-LV-02`	假別選擇器	onChange	更新表單驗證規則	-
`E-LV-03`	日期選擇器	onChange	計算請假天數	GET /api/v1/leave/calculate-days
`E-LV-04`	附件上傳	onUpload	上傳附件	POST /api/v1/files/upload
`E-LV-05`	儲存草稿	onClick	儲存至 LocalStorage	-
`E-LV-06`	送出申請	onClick	提交請假申請	POST /api/v1/leave/applications

E-LV-06 詳細流程:

```
const handleSubmitLeave = async (values: LeaveFormData) => {
  try {
    // 1. 表單驗證
    await form.validateFields();

    // 2. 餘額檢查
    const selectedType = leaveTypes.find(t => t.leaveTypeId === values.leaveTypeId);
    const balance = balances.find(b => b.leaveTypeId === values.leaveTypeId);

    if (balance && values.totalDays > balance.remainingDays) {
      message.error(`剩餘餘額不足，目前剩餘 ${balance.remainingDays} 天`);
      return;
    }

    // 3. 組裝請求
    const request: CreateLeaveRequest = {
      employeeId: currentUser.employeeId,
      leaveTypeId: values.leaveTypeId,
      startDate: values.dateRange[0].format('YYYY-MM-DD'),
      endDate: values.dateRange[1].format('YYYY-MM-DD'),
      startPeriod: values.startPeriod,
      endPeriod: values.endPeriod,
      reason: values.reason,
      proofAttachmentUrl: values.attachmentUrl
    };

    // 4. 提交申請
    setSubmitting(true);
    const response = await leaveService.createApplication(request);

    // 5. 更新Redux狀態
    dispatch(addLeaveApplication(response));
    dispatch(refreshLeaveBalances());

    // 6. 顯示成功
    message.success('請假申請已送出，待主管審核');
    form.resetFields();

  } catch (error) {
    message.error('申請失敗: ' + error.message);
  } finally {
    setSubmitting(false);
  }
};
```

4.4 差勤審核頁面事件 (HR03-P06)

事件ID	觸發元素	事件類型	事件處理	後端API
`E-APR-01`	Tab切換	onChange	載入對應類型的待審清單	GET /api/v1/attendance/approvals?type={type}
`E-APR-02`	申請項目點擊	onClick	顯示申請詳情	-
`E-APR-03`	核准按鈕	onClick	確認核准	PUT /api/v1/leave/applications/{id}/approve
`E-APR-04`	駁回按鈕	onClick	開啟駁回原因對話框	-
`E-APR-05`	駁回確認	onClick	執行駁回	PUT /api/v1/leave/applications/{id}/reject
`E-APR-06`	批次核准	onClick	批次核准選中項目	PUT /api/v1/leave/applications/batch-approve

5. Data Flow設計

5.1 前端狀態管理 (Redux)

5.1.1 State結構

```
interface AttendanceState {
  // 打卡相關
  checkIn: {
    todayRecord: AttendanceRecord | null;
    shift: Shift | null;
    loading: boolean;
  };

  // 差勤記錄
  records: {
    list: AttendanceRecord[];
    selectedMonth: string;
    statistics: MonthlyStatistics | null;
    loading: boolean;
  };

  // 請假相關
  leave: {
    balances: LeaveBalance[];
    leaveTypes: LeaveType[];
    applications: LeaveApplication[];
    loading: boolean;
  };

  // 加班相關
  overtime: {
    applications: OvertimeApplication[];
    monthlyStatistics: OvertimeStatistics | null;
    loading: boolean;
  };

  // 審核相關 (主管)
  approvals: {
    pendingLeaves: LeaveApplication[];
    pendingOvertimes: OvertimeApplication[];
    pendingCorrections: CorrectionRequest[];
    loading: boolean;
  };
}

interface MonthlyStatistics {
  month: string;
  totalWorkDays: number;
  actualWorkDays: number;
```

```
  totalLeaveDays: number;
  totalOvertimeHours: number;
  lateCount: number;
  earlyLeaveCount: number;
}
```

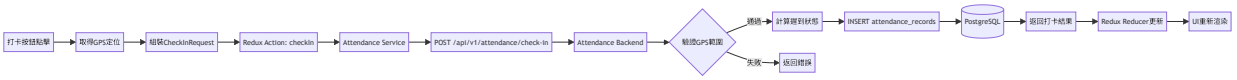
5.1.2 Redux Actions

```
// 打卡相關Actions
export const attendanceActions = {
  loadTodayRecord: createAsyncThunk(
    'attendance/loadToday',
    async (employeeId: string) => {
      const response = await attendanceService.getTodayRecord(employeeId);
      return response;
    }
  ),

  checkIn: createAsyncThunk(
    'attendance/checkIn',
    async (request: CheckInRequest) => {
      const response = await attendanceService.checkIn(request);
      return response;
    }
  ),

  checkOut: createAsyncThunk(
    'attendance/checkOut',
    async (request: CheckOutRequest) => {
      const response = await attendanceService.checkOut(request);
      return response;
    }
  ),
};

// 請假相關Actions
export const leaveActions = {
  loadBalances: createAsyncThunk(
    'leave/loadBalances',
    async (employeeId: string) => {
      const response = await leaveService.getBalances(employeeId);
      return response;
    }
  ),

  createApplication: createAsyncThunk(
    'leave/createApplication',
    async (request: CreateLeaveRequest) => {
      const response = await leaveService.createApplication(request);
      return response;
    }
  ),

  approveLeave: createAsyncThunk(
    'leave/approve',
    async (applicationId: string) => {
      await leaveService.approveApplication(applicationId);
```

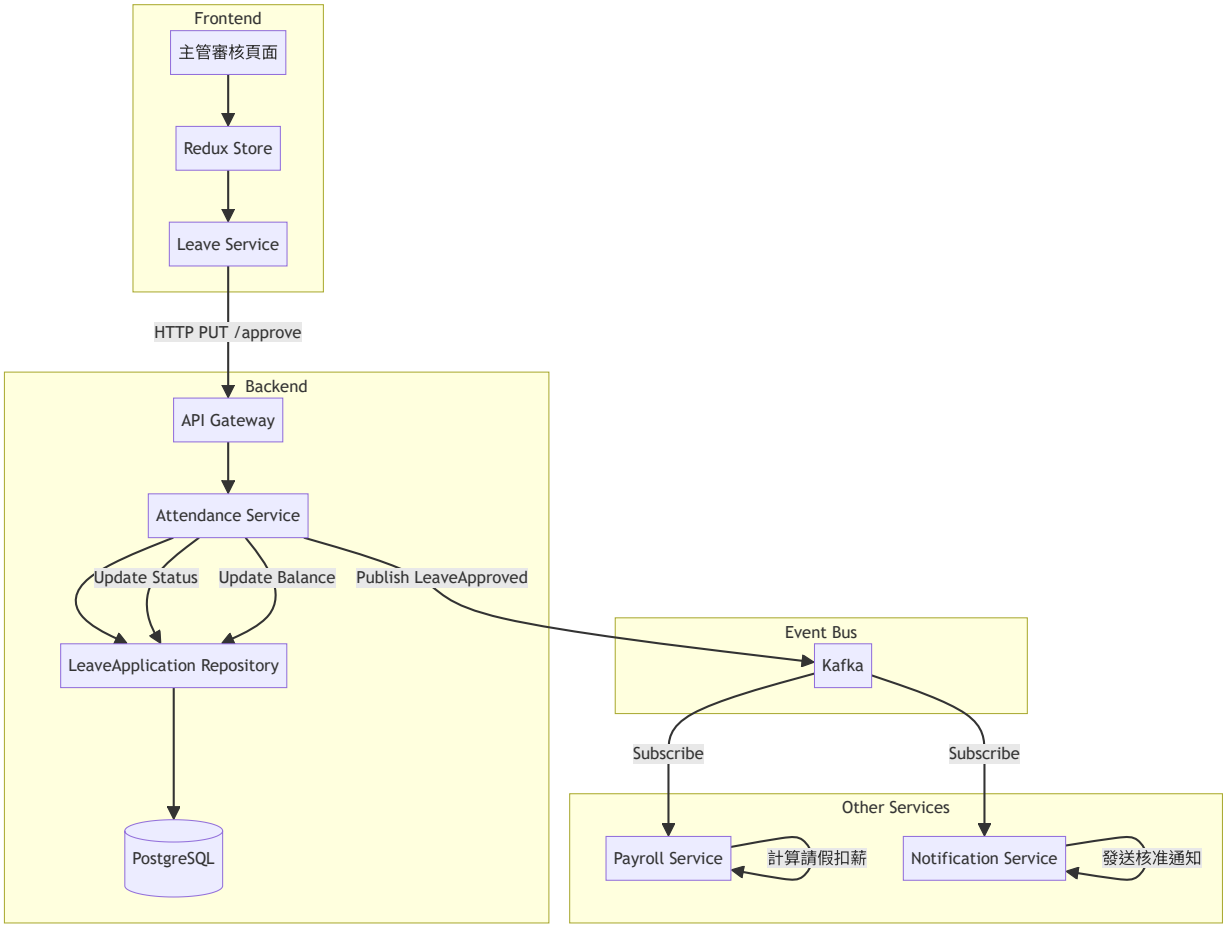
```
      return applicationId;
    }
  ),
};
```

5.2 前後端資料流

5.2.1 打卡流程資料流

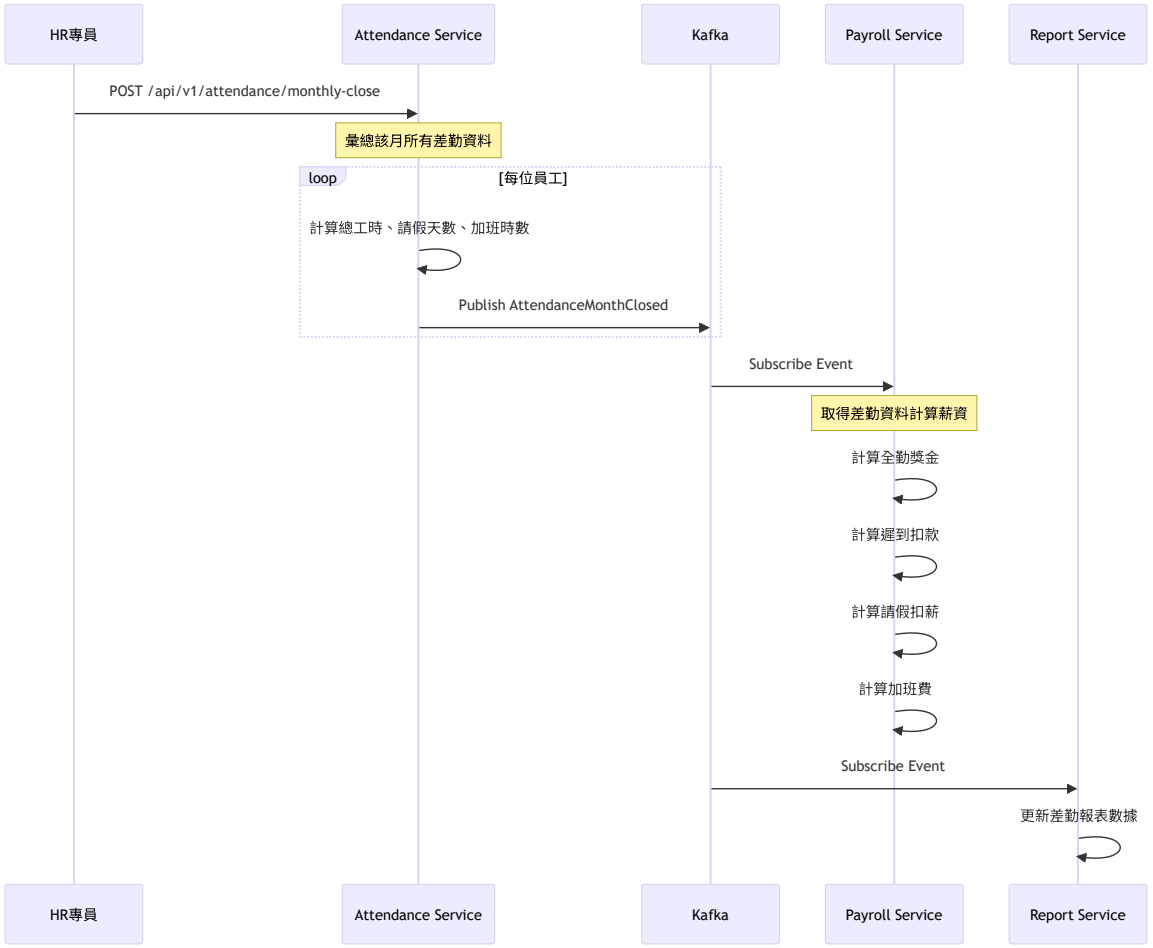


5.2.2 請假審核資料流



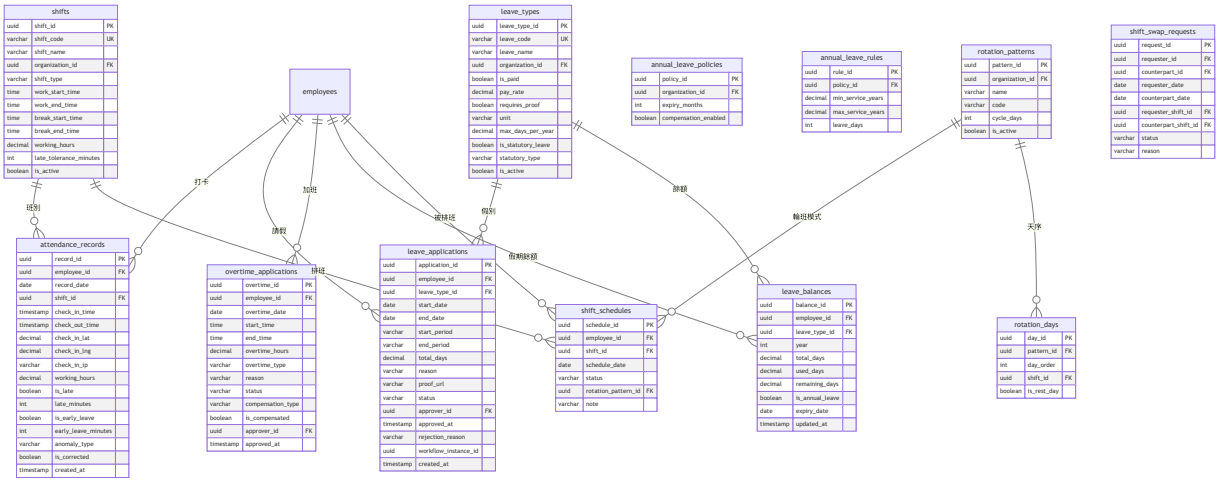
5.3 服務間資料流

5.3.1 月度差勤結算事件流



6. 資料庫設計

6.1 ER Diagram



6.2 DDL Script

```
-- 班別表
CREATE TABLE shifts (
    shift_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    shift_code VARCHAR(50) UNIQUE NOT NULL,
    shift_name VARCHAR(255) NOT NULL,
    organization_id UUID NOT NULL,
    shift_type VARCHAR(20) NOT NULL CHECK (shift_type IN ('STANDARD', 'FLEXIBLE', 'ROTATING
    work_start_time TIME NOT NULL,
    work_end_time TIME NOT NULL,
    break_start_time TIME,
    break_end_time TIME,
    working_hours DECIMAL(4,2) NOT NULL,
    late_tolerance_minutes INTEGER DEFAULT 0,
    early_leave_tolerance_minutes INTEGER DEFAULT 0,
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_shifts_org ON shifts(organization_id);
CREATE INDEX idx_shifts_active ON shifts(is_active) WHERE is_active = TRUE;

COMMENT ON TABLE shifts IS '班別設定表';
COMMENT ON COLUMN shifts.shift_type IS '班別類型: STANDARD標準班, FLEXIBLE彈性班, ROTATING輪班
COMMENT ON COLUMN shifts.late_tolerance_minutes IS '遲到容許分鐘數';

-- 假別表
CREATE TABLE leave_types (
    leave_type_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    leave_code VARCHAR(50) UNIQUE NOT NULL,
    leave_name VARCHAR(255) NOT NULL,
    organization_id UUID,
    is_paid BOOLEAN DEFAULT FALSE,
    pay_rate DECIMAL(3,2) DEFAULT 0 CHECK (pay_rate >= 0 AND pay_rate <= 1),
    requires_proof BOOLEAN DEFAULT FALSE,
    proof_description TEXT,
    unit VARCHAR(20) NOT NULL CHECK (unit IN ('HOUR', 'HALF_DAY', 'FULL_DAY')),
    max_days_per_year DECIMAL(5,2),
    can_carryover BOOLEAN DEFAULT FALSE,
    is_statutory_leave BOOLEAN DEFAULT FALSE,
    statutory_type VARCHAR(50),
    is_active BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_leave_types_org ON leave_types(organization_id);
CREATE INDEX idx_leave_types_statutory ON leave_types(is_statutory_leave);

COMMENT ON TABLE leave_types IS '假別設定表';
```

```
COMMENT ON COLUMN leave_types.is_paid IS '是否支薪';
COMMENT ON COLUMN leave_types.pay_rate IS '支薪比例 (1=全薪, 0.5=半薪)';
COMMENT ON COLUMN leave_types.statutory_type IS '法定假別類型: ANNUAL_LEAVE, SICK_LEAVE, MAR

-- 打卡記錄表
CREATE TABLE attendance_records (
    record_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    employee_id UUID NOT NULL,
    record_date DATE NOT NULL,
    shift_id UUID REFERENCES shifts(shift_id),
    check_in_time TIMESTAMP,
    check_out_time TIMESTAMP,
    check_in_latitude DECIMAL(10,7),
    check_in_longitude DECIMAL(10,7),
    check_in_ip VARCHAR(50),
    check_out_latitude DECIMAL(10,7),
    check_out_longitude DECIMAL(10,7),
    check_out_ip VARCHAR(50),
    working_hours DECIMAL(5,2),
    is_late BOOLEAN DEFAULT FALSE,
    late_minutes INTEGER DEFAULT 0,
    is_early_leave BOOLEAN DEFAULT FALSE,
    early_leave_minutes INTEGER DEFAULT 0,
    anomaly_type VARCHAR(30) CHECK (anomaly_type IN ('LATE', 'EARLY_LEAVE', 'MISSING_CHECK_
    anomaly_note TEXT,
    is_corrected BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT uk_attendance_employee_date UNIQUE (employee_id, record_date)
);

CREATE INDEX idx_attendance_employee ON attendance_records(employee_id);
CREATE INDEX idx_attendance_date ON attendance_records(record_date);
CREATE INDEX idx_attendance_anomaly ON attendance_records(anomaly_type) WHERE anomaly_type

COMMENT ON TABLE attendance_records IS '打卡記錄表';
COMMENT ON COLUMN attendance_records.anomaly_type IS '異常類型: LATE遲到, EARLY_LEAVE早退, M

-- 請假申請表
CREATE TABLE leave_applications (
    application_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    employee_id UUID NOT NULL,
    leave_type_id UUID NOT NULL REFERENCES leave_types(leave_type_id),
    start_date DATE NOT NULL,
    end_date DATE NOT NULL,
    start_period VARCHAR(10) NOT NULL CHECK (start_period IN ('AM', 'PM', 'FULL_DAY')),
    end_period VARCHAR(10) NOT NULL CHECK (end_period IN ('AM', 'PM', 'FULL_DAY')),
    total_days DECIMAL(4,1) NOT NULL,
    total_hours DECIMAL(5,1),
    reason TEXT NOT NULL,
```

```sql
    proof_attachment_url VARCHAR(500),
    applied_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    status VARCHAR(20) NOT NULL DEFAULT 'PENDING' CHECK (status IN ('DRAFT', 'PENDING', 'AP
    approver_id UUID,
    approved_at TIMESTAMP,
    rejection_reason TEXT,
    workflow_instance_id UUID,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT chk_leave_dates CHECK (end_date >= start_date)
);

CREATE INDEX idx_leave_app_employee ON leave_applications(employee_id);
CREATE INDEX idx_leave_app_status ON leave_applications(status);
CREATE INDEX idx_leave_app_dates ON leave_applications(start_date, end_date);
CREATE INDEX idx_leave_app_approver ON leave_applications(approver_id) WHERE status = 'PEND

COMMENT ON TABLE leave_applications IS '請假申請表';
COMMENT ON COLUMN leave_applications.start_period IS '起始時段: AM上午, PM下午, FULL_DAY全天'

-- 加班申請表
CREATE TABLE overtime_applications (
    overtime_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    employee_id UUID NOT NULL,
    overtime_date DATE NOT NULL,
    start_time TIME NOT NULL,
    end_time TIME NOT NULL,
    overtime_hours DECIMAL(4,1) NOT NULL,
    overtime_type VARCHAR(20) NOT NULL CHECK (overtime_type IN ('WEEKDAY', 'REST_DAY', 'HOL
    reason TEXT NOT NULL,
    applied_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    status VARCHAR(20) NOT NULL DEFAULT 'PENDING' CHECK (status IN ('DRAFT', 'PENDING', 'AP
    approver_id UUID,
    approved_at TIMESTAMP,
    rejection_reason TEXT,
    compensation_type VARCHAR(20) CHECK (compensation_type IN ('PAY', 'COMP_TIME')),
    is_compensated BOOLEAN DEFAULT FALSE,
    workflow_instance_id UUID,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT chk_overtime_times CHECK (end_time > start_time)
);

CREATE INDEX idx_overtime_employee ON overtime_applications(employee_id);
CREATE INDEX idx_overtime_date ON overtime_applications(overtime_date);
CREATE INDEX idx_overtime_status ON overtime_applications(status);

COMMENT ON TABLE overtime_applications IS '加班申請表';
COMMENT ON COLUMN overtime_applications.overtime_type IS '加班類型: WEEKDAY平日, REST_DAY休息
```

```sql
COMMENT ON COLUMN overtime_applications.compensation_type IS '補償方式: PAY加班費, COMP_TIME

-- 假期餘額表
CREATE TABLE leave_balances (
    balance_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    employee_id UUID NOT NULL,
    leave_type_id UUID NOT NULL REFERENCES leave_types(leave_type_id),
    year INTEGER NOT NULL,
    total_days DECIMAL(5,1) NOT NULL,
    used_days DECIMAL(5,1) NOT NULL DEFAULT 0,
    carry_over_days DECIMAL(5,1) NOT NULL DEFAULT 0,
    is_annual_leave BOOLEAN DEFAULT FALSE,
    expiry_date DATE,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT uk_balance_emp_type_year UNIQUE (employee_id, leave_type_id, year)
);

CREATE INDEX idx_balance_employee ON leave_balances(employee_id);
CREATE INDEX idx_balance_expiry ON leave_balances(expiry_date) WHERE is_annual_leave = TRUE

COMMENT ON TABLE leave_balances IS '假期餘額表';
COMMENT ON COLUMN leave_balances.carry_over_days IS '上年度結轉天數';
COMMENT ON COLUMN leave_balances.expiry_date IS '特休假到期日';

-- 特休假政策表
CREATE TABLE annual_leave_policies (
    policy_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    organization_id UUID NOT NULL UNIQUE,
    annual_leave_system VARCHAR(20) NOT NULL DEFAULT 'CALENDAR_YEAR',
    overdraw_policy VARCHAR(30) NOT NULL DEFAULT 'DENY',
    expiry_policy VARCHAR(30) NOT NULL DEFAULT 'PAY_COMPENSATION',
    carry_over_limit INTEGER,
    expiry_months INTEGER DEFAULT 12,
    compensation_enabled BOOLEAN DEFAULT TRUE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

COMMENT ON COLUMN annual_leave_policies.annual_leave_system IS '年度制度: CALENDAR_YEAR歷年制
COMMENT ON COLUMN annual_leave_policies.overdraw_policy IS '超額請假政策: DENY拒絕, CONVERT_
COMMENT ON COLUMN annual_leave_policies.expiry_policy IS '未休假處理: CARRYOVER結轉, PAY_COM
COMMENT ON COLUMN annual_leave_policies.carry_over_limit IS '結轉上限天數 (CARRYOVER政策使用)

-- 特休假規則表 (年資對應天數)
CREATE TABLE annual_leave_rules (
    rule_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    policy_id UUID NOT NULL REFERENCES annual_leave_policies(policy_id),
    min_service_years DECIMAL(4,2) NOT NULL,
    max_service_years DECIMAL(4,2),
    leave_days INTEGER NOT NULL,
```

```sql
    CONSTRAINT uk_rule_policy_years UNIQUE (policy_id, min_service_years)
);

COMMENT ON TABLE annual_leave_rules IS '特休假規則表 (依勞基法年資對應天數)';

-- 補卡申請表
CREATE TABLE attendance_corrections (
    correction_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    record_id UUID NOT NULL REFERENCES attendance_records(record_id),
    employee_id UUID NOT NULL,
    correction_type VARCHAR(20) NOT NULL CHECK (correction_type IN ('CHECK_IN', 'CHECK_OUT'
    original_time TIMESTAMP,
    corrected_time TIMESTAMP NOT NULL,
    reason TEXT NOT NULL,
    status VARCHAR(20) NOT NULL DEFAULT 'PENDING' CHECK (status IN ('PENDING', 'APPROVED',
    approver_id UUID,
    approved_at TIMESTAMP,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_correction_employee ON attendance_corrections(employee_id);
CREATE INDEX idx_correction_status ON attendance_corrections(status);

COMMENT ON TABLE attendance_corrections IS '補卡申請表';

-- 排班表
CREATE TABLE shift_schedules (
    schedule_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    employee_id UUID NOT NULL,
    shift_id UUID NOT NULL REFERENCES shifts(shift_id),
    schedule_date DATE NOT NULL,
    status VARCHAR(20) NOT NULL DEFAULT 'DRAFT' CHECK (status IN ('DRAFT', 'PUBLISHED', 'LO
    rotation_pattern_id UUID,
    note TEXT,
    is_deleted BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT uk_schedule_emp_date UNIQUE (employee_id, schedule_date)
);

CREATE INDEX idx_schedule_employee ON shift_schedules(employee_id);
CREATE INDEX idx_schedule_date ON shift_schedules(schedule_date);

COMMENT ON TABLE shift_schedules IS '排班表';
COMMENT ON COLUMN shift_schedules.status IS '排班狀態: DRAFT草稿, PUBLISHED已發佈, LOCKED已鎖

-- 輪班模式表
CREATE TABLE rotation_patterns (
    pattern_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    organization_id UUID NOT NULL,
```

```sql
    name VARCHAR(100) NOT NULL,
    code VARCHAR(50) NOT NULL,
    cycle_days INTEGER NOT NULL,
    is_active BOOLEAN DEFAULT TRUE,
    is_deleted BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_rotation_org ON rotation_patterns(organization_id);

COMMENT ON TABLE rotation_patterns IS '輪班模式表';
COMMENT ON COLUMN rotation_patterns.cycle_days IS '一個完整循環的天數';

-- 輪班天序表
CREATE TABLE rotation_days (
    day_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    pattern_id UUID NOT NULL REFERENCES rotation_patterns(pattern_id),
    day_order INTEGER NOT NULL,
    shift_id UUID REFERENCES shifts(shift_id),
    is_rest_day BOOLEAN DEFAULT FALSE,

    CONSTRAINT uk_rotation_day_order UNIQUE (pattern_id, day_order)
);

COMMENT ON TABLE rotation_days IS '輪班天序表（循環中每天的班別）';

-- 換班申請表
CREATE TABLE shift_swap_requests (
    request_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    requester_id UUID NOT NULL,
    counterpart_id UUID NOT NULL,
    requester_date DATE NOT NULL,
    counterpart_date DATE NOT NULL,
    requester_shift_id UUID REFERENCES shifts(shift_id),
    counterpart_shift_id UUID REFERENCES shifts(shift_id),
    status VARCHAR(30) NOT NULL DEFAULT 'PENDING_COUNTERPART'
        CHECK (status IN ('PENDING_COUNTERPART', 'PENDING_APPROVAL', 'APPROVED', 'REJECTED'
    reason TEXT,
    rejection_reason TEXT,
    approver_id UUID,
    is_deleted BOOLEAN DEFAULT FALSE,
    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX idx_swap_requester ON shift_swap_requests(requester_id);
CREATE INDEX idx_swap_counterpart ON shift_swap_requests(counterpart_id);
CREATE INDEX idx_swap_status ON shift_swap_requests(status);
```

```
COMMENT ON TABLE shift_swap_requests IS '換班申請表';
COMMENT ON COLUMN shift_swap_requests.status IS '換班狀態: PENDING_COUNTERPART等待對方同意, P
```

6.3 資料字典

表名	欄位	類型	說明	備註
`shifts`	shift_type	ENUM	班別類型	STANDARD/FLEXIBLE/ROTATING
`leave_types`	unit	ENUM	請假單位	HOUR/HALF_DAY/FULL_DAY
`leave_types`	statutory_type	ENUM	法定假別	ANNUAL_LEAVE, SICK_LEAVE等
`attendance_records`	anomaly_type	ENUM	異常類型	LATE, EARLY_LEAVE等
`leave_applications`	status	ENUM	申請狀態	DRAFT/PENDING/APPROVED/REJECTED/CANCELLED
`overtime_applications`	overtime_type	ENUM	加班類型	WEEKDAY/REST_DAY/HOLIDAY
`overtime_applications`	compensation_type	ENUM	補償方式	PAY/COMP_TIME

6.4 初始化資料

```
-- 初始化法定假別
INSERT INTO leave_types (leave_code, leave_name, is_paid, pay_rate, requires_proof, unit, m
('ANNUAL', '特休假', TRUE, 1.0, FALSE, 'FULL_DAY', NULL, TRUE, 'ANNUAL_LEAVE'),
('SICK', '病假', FALSE, 0.5, TRUE, 'HALF_DAY', 30, TRUE, 'SICK_LEAVE'),
('PERSONAL', '事假', FALSE, 0.0, FALSE, 'HALF_DAY', 14, TRUE, 'PERSONAL_LEAVE'),
('MARRIAGE', '婚假', TRUE, 1.0, TRUE, 'FULL_DAY', 8, TRUE, 'MARRIAGE_LEAVE'),
('BEREAVEMENT_PARENT', '喪假(父母)', TRUE, 1.0, TRUE, 'FULL_DAY', 8, TRUE, 'BEREAVEMENT_LEA
('BEREAVEMENT_SPOUSE', '喪假(配偶)', TRUE, 1.0, TRUE, 'FULL_DAY', 8, TRUE, 'BEREAVEMENT_LEA
('BEREAVEMENT_CHILD', '喪假(子女)', TRUE, 1.0, TRUE, 'FULL_DAY', 8, TRUE, 'BEREAVEMENT_LEAVE
('MATERNITY', '產假', TRUE, 1.0, TRUE, 'FULL_DAY', 56, TRUE, 'MATERNITY_LEAVE'),
('PATERNITY', '陪產假', TRUE, 1.0, TRUE, 'FULL_DAY', 7, TRUE, 'PATERNITY_LEAVE'),
('MENSTRUAL', '生理假', FALSE, 0.5, FALSE, 'FULL_DAY', 12, TRUE, 'MENSTRUAL_LEAVE'),
('PARENTAL', '育嬰留停', FALSE, 0.0, TRUE, 'FULL_DAY', NULL, TRUE, 'PARENTAL_LEAVE'),
('COMP_TIME', '補休', TRUE, 1.0, FALSE, 'HOUR', NULL, FALSE, NULL);

-- 初始化標準班別
INSERT INTO shifts (shift_code, shift_name, organization_id, shift_type, work_start_time, w
('STD', '標準班', '00000000-0000-0000-0000-000000000001', 'STANDARD', '09:00', '18:00', '12
('FLEX_A', '彈性班A(08:00-17:00)', '00000000-0000-0000-0000-000000000001', 'FLEXIBLE', '08:0
('FLEX_B', '彈性班B(10:00-19:00)', '00000000-0000-0000-0000-000000000001', 'FLEXIBLE', '10:0

-- 初始化特休假規則（勞基法）
INSERT INTO annual_leave_policies (organization_id, expiry_months, compensation_enabled) VA
('00000000-0000-0000-0000-000000000001', 12, TRUE);

INSERT INTO annual_leave_rules (policy_id, min_service_years, max_service_years, leave_days
((SELECT policy_id FROM annual_leave_policies LIMIT 1), 0.5, 1, 3),
((SELECT policy_id FROM annual_leave_policies LIMIT 1), 1, 2, 7),
((SELECT policy_id FROM annual_leave_policies LIMIT 1), 2, 3, 10),
((SELECT policy_id FROM annual_leave_policies LIMIT 1), 3, 5, 14),
((SELECT policy_id FROM annual_leave_policies LIMIT 1), 5, 10, 15),
((SELECT policy_id FROM annual_leave_policies LIMIT 1), 10, NULL, 15);  -- 10年以上每年+1天
```

(文件持續，下一部分包含Domain設計、領域事件設計、完整API規格等)

Continued: 03_考勤管理服務系統設計書_part3.md

7. Domain設計

7.1 聚合根 (Aggregate Root)

7.1.1 Shift聚合根 (班別)

職責: 定義工作班別與工時規則

Java實作:

```java
@Entity
@Table(name = "shifts")
public class Shift {
    @EmbeddedId
    private ShiftId id;

    @Column(name = "shift_code", unique = true, nullable = false)
    private String shiftCode;

    @Column(name = "shift_name", nullable = false)
    private String shiftName;

    @Column(name = "organization_id", nullable = false)
    private UUID organizationId;

    @Enumerated(EnumType.STRING)
    @Column(name = "shift_type", nullable = false)
    private ShiftType shiftType;

    @Embedded
    private WorkingTime workingTime;

    @Column(name = "late_tolerance_minutes")
    private Integer lateToleranceMinutes;

    @Column(name = "early_leave_tolerance_minutes")
    private Integer earlyLeaveToleranceMinutes;

    @Column(name = "is_active")
    private boolean isActive;

    // ========== Domain行為 ==========

    /**
     * 檢查打卡時間是否遲到
     */
    public LateResult checkLate(LocalTime checkInTime) {
        LocalTime tolerance = workingTime.getWorkStartTime()
            .plusMinutes(lateToleranceMinutes);

        if (checkInTime.isAfter(tolerance)) {
            int lateMinutes = (int) Duration.between(
                workingTime.getWorkStartTime(), checkInTime).toMinutes();
            return new LateResult(true, lateMinutes);
        }
        return new LateResult(false, 0);
    }

    /**
     * 檢查打卡時間是否早退
     */
```

```java
    public EarlyLeaveResult checkEarlyLeave(LocalTime checkOutTime) {
        LocalTime tolerance = workingTime.getWorkEndTime()
            .minusMinutes(earlyLeaveToleranceMinutes);

        if (checkOutTime.isBefore(tolerance)) {
            int earlyMinutes = (int) Duration.between(
                checkOutTime, workingTime.getWorkEndTime()).toMinutes();
            return new EarlyLeaveResult(true, earlyMinutes);
        }
        return new EarlyLeaveResult(false, 0);
    }

    /**
     * 計算實際工時
     */
    public BigDecimal calculateWorkingHours(LocalTime checkIn, LocalTime checkOut) {
        Duration workDuration = Duration.between(checkIn, checkOut);

        // 扣除休息時間
        if (hasBreakTime()) {
            Duration breakDuration = Duration.between(
                workingTime.getBreakStartTime(),
                workingTime.getBreakEndTime());
            workDuration = workDuration.minus(breakDuration);
        }

        return BigDecimal.valueOf(workDuration.toMinutes())
            .divide(BigDecimal.valueOf(60), 2, RoundingMode.HALF_UP);
    }
}

// 值對象
@Embeddable
public class WorkingTime {
    @Column(name = "work_start_time")
    private LocalTime workStartTime;

    @Column(name = "work_end_time")
    private LocalTime workEndTime;

    @Column(name = "break_start_time")
    private LocalTime breakStartTime;

    @Column(name = "break_end_time")
    private LocalTime breakEndTime;
}
```

7.1.2 AttendanceRecord聚合根 (打卡記錄)

職責: 記錄員工打卡與計算出勤狀態

Java實作:

```java
@Entity
@Table(name = "attendance_records")
public class AttendanceRecord {
    @EmbeddedId
    private RecordId id;

    @Column(name = "employee_id", nullable = false)
    private UUID employeeId;

    @Column(name = "record_date", nullable = false)
    private LocalDate recordDate;

    @Column(name = "shift_id")
    private UUID shiftId;

    @Embedded
    private CheckInInfo checkInInfo;

    @Embedded
    private CheckOutInfo checkOutInfo;

    @Column(name = "working_hours")
    private BigDecimal workingHours;

    @Column(name = "is_late")
    private boolean isLate;

    @Column(name = "late_minutes")
    private int lateMinutes;

    @Column(name = "is_early_leave")
    private boolean isEarlyLeave;

    @Column(name = "early_leave_minutes")
    private int earlyLeaveMinutes;

    @Enumerated(EnumType.STRING)
    @Column(name = "anomaly_type")
    private AnomalyType anomalyType;

    @Column(name = "is_corrected")
    private boolean isCorrected;

    // ========== Factory Method ==========

    /**
     * 上班打卡
     */
    public static AttendanceRecord checkIn(
            UUID employeeId,
            LocalDate date,
```

```
            Shift shift,
            CheckInRequest request) {

        AttendanceRecord record = new AttendanceRecord();
        record.id = RecordId.generate();
        record.employeeId = employeeId;
        record.recordDate = date;
        record.shiftId = shift.getId().getValue();

        // 設定打卡資訊
        record.checkInInfo = new CheckInInfo(
            request.getCheckInTime(),
            request.getLatitude(),
            request.getLongitude(),
            request.getIpAddress()
        );

        // 檢查遲到
        LateResult lateResult = shift.checkLate(
            request.getCheckInTime().toLocalTime());
        record.isLate = lateResult.isLate();
        record.lateMinutes = lateResult.getLateMinutes();

        if (record.isLate) {
            record.anomalyType = AnomalyType.LATE;

            // 發布異常事件
            DomainEventPublisher.publish(new AttendanceAnomalyDetectedEvent(
                record.employeeId,
                record.recordDate,
                AnomalyType.LATE,
                record.lateMinutes
            ));
        }

        // 發布打卡事件
        DomainEventPublisher.publish(new AttendanceRecordedEvent(
            record.id.getValue(),
            record.employeeId,
            record.recordDate,
            record.checkInInfo.getCheckInTime(),
            null
        ));

        return record;
    }

    /**
     * 下班打卡
     */
    public void checkOut(Shift shift, CheckOutRequest request) {
```

```java
        if (this.checkOutInfo != null && this.checkOutInfo.getCheckOutTime() != null) {
            throw new DomainException("今日已完成下班打卡");
        }

        this.checkOutInfo = new CheckOutInfo(
            request.getCheckOutTime(),
            request.getLatitude(),
            request.getLongitude(),
            request.getIpAddress()
        );

        // 檢查早退
        EarlyLeaveResult result = shift.checkEarlyLeave(
            request.getCheckOutTime().toLocalTime());
        this.isEarlyLeave = result.isEarlyLeave();
        this.earlyLeaveMinutes = result.getEarlyMinutes();

        if (this.isEarlyLeave) {
            this.anomalyType = AnomalyType.EARLY_LEAVE;
        }

        // 計算工時
        this.workingHours = shift.calculateWorkingHours(
            this.checkInInfo.getCheckInTime().toLocalTime(),
            request.getCheckOutTime().toLocalTime()
        );
    }

    /**
     * 補卡
     */
    public void correct(LocalDateTime correctedTime, CorrectionType type) {
        if (type == CorrectionType.CHECK_IN) {
            this.checkInInfo = new CheckInInfo(
                correctedTime, null, null, "CORRECTED");
            this.isLate = false;
            this.lateMinutes = 0;
        } else {
            this.checkOutInfo = new CheckOutInfo(
                correctedTime, null, null, "CORRECTED");
            this.isEarlyLeave = false;
            this.earlyLeaveMinutes = 0;
        }
        this.isCorrected = true;
        this.anomalyType = null;
    }
}

// 值對象
@Embeddable
public class CheckInInfo {
```

```
    @Column(name = "check_in_time")
    private LocalDateTime checkInTime;

    @Column(name = "check_in_latitude")
    private BigDecimal latitude;

    @Column(name = "check_in_longitude")
    private BigDecimal longitude;

    @Column(name = "check_in_ip")
    private String ipAddress;
}
```

7.1.3 LeaveApplication聚合根 (請假申請)

職責: 管理員工請假申請與審核

Java實作:

```java
@Entity
@Table(name = "leave_applications")
public class LeaveApplication {
    @EmbeddedId
    private ApplicationId id;

    @Column(name = "employee_id", nullable = false)
    private UUID employeeId;

    @Column(name = "leave_type_id", nullable = false)
    private UUID leaveTypeId;

    @Embedded
    private LeavePeriod leavePeriod;

    @Column(name = "total_days", nullable = false)
    private BigDecimal totalDays;

    @Column(name = "reason", nullable = false)
    private String reason;

    @Column(name = "proof_attachment_url")
    private String proofAttachmentUrl;

    @Enumerated(EnumType.STRING)
    @Column(name = "status", nullable = false)
    private ApplicationStatus status;

    @Column(name = "approver_id")
    private UUID approverId;

    @Column(name = "approved_at")
    private LocalDateTime approvedAt;

    @Column(name = "rejection_reason")
    private String rejectionReason;

    // ========== Factory Method ==========

    /**
     * 建立請假申請
     */
    public static LeaveApplication create(
            UUID employeeId,
            LeaveType leaveType,
            LeaveBalance balance,
            CreateLeaveRequest request) {

        // 計算請假天數
        BigDecimal days = calculateLeaveDays(
            request.getStartDate(),
```

```java
            request.getEndDate(),
            request.getStartPeriod(),
            request.getEndPeriod()
        );

        // 驗證餘額
        if (balance.getRemainingDays().compareTo(days) < 0) {
            throw new DomainException(
                String.format("假期餘額不足，剩餘 %s 天，申請 %s 天",
                    balance.getRemainingDays(), days));
        }

        // 驗證是否需要證明文件
        if (leaveType.isRequiresProof() &&
            StringUtils.isBlank(request.getProofAttachmentUrl())) {
            throw new DomainException("此假別需要上傳證明文件");
        }

        LeaveApplication app = new LeaveApplication();
        app.id = ApplicationId.generate();
        app.employeeId = employeeId;
        app.leaveTypeId = leaveType.getId().getValue();
        app.leavePeriod = new LeavePeriod(
            request.getStartDate(),
            request.getEndDate(),
            request.getStartPeriod(),
            request.getEndPeriod()
        );
        app.totalDays = days;
        app.reason = request.getReason();
        app.proofAttachmentUrl = request.getProofAttachmentUrl();
        app.status = ApplicationStatus.PENDING;

        // 發布事件
        DomainEventPublisher.publish(new LeaveAppliedEvent(
            app.id.getValue(),
            app.employeeId,
            app.leaveTypeId,
            app.totalDays
        ));

        return app;
    }

    /**
     * 核准
     */
    public void approve(UUID approverId, LeaveBalance balance) {
        if (this.status != ApplicationStatus.PENDING) {
            throw new DomainException("只能核准待審核的申請");
        }
```

```
        this.status = ApplicationStatus.APPROVED;
        this.approverId = approverId;
        this.approvedAt = LocalDateTime.now();

        // 扣除餘額
        balance.deduct(this.totalDays);

        // 發布事件
        DomainEventPublisher.publish(new LeaveApprovedEvent(
            this.id.getValue(),
            this.employeeId,
            this.leaveTypeId,
            this.totalDays
        ));
    }

    /**
     * 駁回
     */
    public void reject(UUID approverId, String reason) {
        if (this.status != ApplicationStatus.PENDING) {
            throw new DomainException("只能駁回待審核的申請");
        }

        this.status = ApplicationStatus.REJECTED;
        this.approverId = approverId;
        this.rejectionReason = reason;

        DomainEventPublisher.publish(new LeaveRejectedEvent(
            this.id.getValue(),
            this.employeeId,
            reason
        ));
    }

    /**
     * 取消
     */
    public void cancel(LeaveBalance balance) {
        if (this.status == ApplicationStatus.CANCELLED) {
            throw new DomainException("申請已取消");
        }

        // 若已核准，退回餘額
        if (this.status == ApplicationStatus.APPROVED) {
            balance.refund(this.totalDays);
        }

        this.status = ApplicationStatus.CANCELLED;
```

```
    }
}
```

7.1.4 OvertimeApplication聚合根 (加班申請)

職責: 管理員工加班申請，包含勞基法時數管控

Java實作:

```java
@Entity
@Table(name = "overtime_applications")
public class OvertimeApplication {
    @EmbeddedId
    private OvertimeId id;

    @Column(name = "employee_id", nullable = false)
    private UUID employeeId;

    @Column(name = "overtime_date", nullable = false)
    private LocalDate overtimeDate;

    @Column(name = "start_time", nullable = false)
    private LocalTime startTime;

    @Column(name = "end_time", nullable = false)
    private LocalTime endTime;

    @Column(name = "overtime_hours", nullable = false)
    private BigDecimal overtimeHours;

    @Enumerated(EnumType.STRING)
    @Column(name = "overtime_type", nullable = false)
    private OvertimeType overtimeType;

    @Column(name = "reason", nullable = false)
    private String reason;

    @Enumerated(EnumType.STRING)
    @Column(name = "status", nullable = false)
    private ApplicationStatus status;

    @Enumerated(EnumType.STRING)
    @Column(name = "compensation_type")
    private CompensationType compensationType;

    @Column(name = "is_compensated")
    private boolean isCompensated;

    // 勞基法限制常數
    private static final BigDecimal MONTHLY_LIMIT = new BigDecimal("46");
    private static final BigDecimal QUARTERLY_LIMIT = new BigDecimal("138");

    // ========== Factory Method ==========

    /**
     * 建立加班申請
     */
    public static OvertimeApplication create(
            UUID employeeId,
            OvertimeStatistics statistics,
```

```
            CreateOvertimeRequest request) {

        // 計算加班時數
        BigDecimal hours = calculateHours(request.getStartTime(), request.getEndTime());

        // 檢查月加班上限 (46小時)
        BigDecimal newMonthly = statistics.getMonthlyHours().add(hours);
        if (newMonthly.compareTo(MONTHLY_LIMIT) > 0) {
            DomainEventPublisher.publish(new OvertimeLimitExceededEvent(
                employeeId,
                newMonthly,
                MONTHLY_LIMIT,
                "MONTHLY"
            ));
            throw new DomainException(
                String.format("超過月加班上限，目前累計 %s 小時 + 本次 %s 小時 > 46小時",
                    statistics.getMonthlyHours(), hours));
        }

        // 檢查季加班上限 (138小時)
        BigDecimal newQuarterly = statistics.getQuarterlyHours().add(hours);
        if (newQuarterly.compareTo(QUARTERLY_LIMIT) > 0) {
            DomainEventPublisher.publish(new OvertimeLimitExceededEvent(
                employeeId,
                newQuarterly,
                QUARTERLY_LIMIT,
                "QUARTERLY"
            ));
            throw new DomainException("超過季加班上限 (138小時)");
        }

        OvertimeApplication app = new OvertimeApplication();
        app.id = OvertimeId.generate();
        app.employeeId = employeeId;
        app.overtimeDate = request.getOvertimeDate();
        app.startTime = request.getStartTime();
        app.endTime = request.getEndTime();
        app.overtimeHours = hours;
        app.overtimeType = request.getOvertimeType();
        app.reason = request.getReason();
        app.compensationType = request.getCompensationType();
        app.status = ApplicationStatus.PENDING;

        DomainEventPublisher.publish(new OvertimeAppliedEvent(
            app.id.getValue(),
            app.employeeId,
            app.overtimeHours,
            app.overtimeType
        ));

        return app;
```

```java
    }

    /**
     * 核准
     */
    public void approve(UUID approverId) {
        if (this.status != ApplicationStatus.PENDING) {
            throw new DomainException("只能核准待審核的申請");
        }

        this.status = ApplicationStatus.APPROVED;

        DomainEventPublisher.publish(new OvertimeApprovedEvent(
            this.id.getValue(),
            this.employeeId,
            this.overtimeHours,
            this.overtimeType,
            this.compensationType
        ));
    }
}
```

7.1.5 LeaveBalance聚合根 (假期餘額)

職責: 管理員工各類假別的餘額

Java實作:

```java
@Entity
@Table(name = "leave_balances")
public class LeaveBalance {
    @EmbeddedId
    private BalanceId id;

    @Column(name = "employee_id", nullable = false)
    private UUID employeeId;

    @Column(name = "leave_type_id", nullable = false)
    private UUID leaveTypeId;

    @Column(name = "year", nullable = false)
    private int year;

    @Column(name = "total_days", nullable = false)
    private BigDecimal totalDays;

    @Column(name = "used_days", nullable = false)
    private BigDecimal usedDays;

    @Column(name = "carry_over_days")
    private BigDecimal carryOverDays;

    @Column(name = "expiry_date")
    private LocalDate expiryDate;

    // ========== Domain行為 ==========

    /**
     * 可用天數 = 年度額度 + 結轉天數 - 已使用天數
     */
    public BigDecimal getAvailableDays() {
        return totalDays.add(carryOverDays).subtract(usedDays);
    }

    /**
     * 扣除假期（含超額請假政策）
     * @return 實際可扣除天數
     */
    public BigDecimal deductWithPolicy(BigDecimal days, OverdrawPolicy overdrawPolicy) {
        BigDecimal available = getAvailableDays();
        if (available.compareTo(days) >= 0) {
            this.usedDays = this.usedDays.add(days);
            return days;
        }
        switch (overdrawPolicy) {
            case DENY: throw new DomainException("假期餘額不足");
            case CONVERT_TO_PERSONAL:
                this.usedDays = this.usedDays.add(available);
                return available; // 超出部分由呼叫端轉為事假
```

```java
            case ADVANCE:
                this.usedDays = this.usedDays.add(days);
                return days; // 預支下年度
            default: throw new DomainException("假期餘額不足");
        }
    }

    /**
     * 扣除假期（預設 DENY 政策）
     */
    public void deduct(BigDecimal days) {
        deductWithPolicy(days, OverdrawPolicy.DENY);
    }

    /**
     * 退回假期
     */
    public void restore(BigDecimal days) {
        this.usedDays = this.usedDays.subtract(days);
    }

    /**
     * 年度結算：回傳未休天數供呼叫端依 ExpiryPolicy 處理
     */
    public BigDecimal settleYear(ExpiryPolicy expiryPolicy) {
        BigDecimal unused = getAvailableDays();
        return unused.compareTo(BigDecimal.ZERO) > 0 ? unused : BigDecimal.ZERO;
    }

    /**
     * 設定結轉天數（由上年度結算帶入）
     */
    public void addCarryOver(BigDecimal days) {
        this.carryOverDays = this.carryOverDays.add(days);
    }

    /**
     * 檢查是否即將到期
     */
    public boolean isExpiringSoon(int daysThreshold) {
        if (this.expiryDate == null) return false;

        LocalDate threshold = LocalDate.now().plusDays(daysThreshold);
        return this.expiryDate.isBefore(threshold) ||
               this.expiryDate.isEqual(threshold);
    }
}
```

7.1.6 AnnualLeavePolicy聚合根 (特休假政策)

職責: 依勞基法計算特休天數

Java實作:

```java
@Entity
@Table(name = "annual_leave_policies")
public class AnnualLeavePolicy {
    @EmbeddedId
    private PolicyId id;

    @Column(name = "organization_id", nullable = false)
    private UUID organizationId;

    @OneToMany(cascade = CascadeType.ALL)
    @JoinColumn(name = "policy_id")
    private List<AnnualLeaveRule> rules;

    @Column(name = "expiry_months")
    private int expiryMonths;

    @Column(name = "compensation_enabled")
    private boolean compensationEnabled;

    /**
     * 計算特休天數（依勞基法）
     */
    public int calculateAnnualLeaveDays(LocalDate hireDate, LocalDate currentDate) {
        BigDecimal serviceYears = calculateServiceYears(hireDate, currentDate);

        // 未滿6個月，無特休
        if (serviceYears.compareTo(new BigDecimal("0.5")) < 0) {
            return 0;
        }

        // 找到對應規則
        for (AnnualLeaveRule rule : rules) {
            if (serviceYears.compareTo(rule.getMinServiceYears()) >= 0) {
                if (rule.getMaxServiceYears() == null ||
                    serviceYears.compareTo(rule.getMaxServiceYears()) < 0) {

                    // 10年以上特殊計算
                    if (serviceYears.compareTo(new BigDecimal("10")) >= 0) {
                        int extraYears = serviceYears.intValue() - 10;
                        int extraDays = Math.min(extraYears, 15); // 最多加15天
                        return rule.getLeaveDays() + extraDays;  // 15 + extra, 上限30
                    }

                    return rule.getLeaveDays();
                }
            }
        }

        return 0;
    }
```

```java
    /**
     * 計算未休工資補償
     */
    public BigDecimal calculateUnusedCompensation(
            LeaveBalance balance,
            BigDecimal dailySalary) {
        if (!compensationEnabled) return BigDecimal.ZERO;

        return balance.getRemainingDays().multiply(dailySalary);
    }
}
```

7.2 Repository介面

```
// AttendanceRecord Repository
public interface IAttendanceRecordRepository {
    AttendanceRecord findById(RecordId id);
    AttendanceRecord findByEmployeeAndDate(UUID employeeId, LocalDate date);
    List<AttendanceRecord> findByEmployeeAndMonth(UUID employeeId, YearMonth month);
    List<AttendanceRecord> findAnomalies(UUID employeeId, LocalDate from, LocalDate to);
    void save(AttendanceRecord record);
}

// LeaveApplication Repository
public interface ILeaveApplicationRepository {
    LeaveApplication findById(ApplicationId id);
    List<LeaveApplication> findByEmployeeId(UUID employeeId);
    List<LeaveApplication> findPendingByApprover(UUID approverId);
    List<LeaveApplication> findOverlapping(UUID employeeId, LocalDate start, LocalDate end)
    void save(LeaveApplication application);
}

// OvertimeApplication Repository
public interface IOvertimeApplicationRepository {
    OvertimeApplication findById(OvertimeId id);
    List<OvertimeApplication> findByEmployeeId(UUID employeeId);
    OvertimeStatistics getStatistics(UUID employeeId, YearMonth month);
    void save(OvertimeApplication application);
}

// LeaveBalance Repository
public interface ILeaveBalanceRepository {
    LeaveBalance findByEmployeeAndTypeAndYear(UUID employeeId, UUID leaveTypeId, int year);
    List<LeaveBalance> findByEmployeeId(UUID employeeId);
    List<LeaveBalance> findExpiringBefore(LocalDate date);
    void save(LeaveBalance balance);
}

// ShiftSchedule Repository
public interface IShiftScheduleRepository {
    void save(ShiftSchedule schedule);
    Optional<ShiftSchedule> findById(ScheduleId id);
    List<ShiftSchedule> findByEmployeeIdAndDateRange(String employeeId, LocalDate startDate
    List<ShiftSchedule> findByDate(LocalDate date);
    void delete(ScheduleId id);
}

// RotationPattern Repository
public interface IRotationPatternRepository {
    void save(RotationPattern pattern);
    Optional<RotationPattern> findById(RotationPatternId id);
    List<RotationPattern> findAll();
    void delete(RotationPatternId id);
```

```
}

// ShiftSwapRequest Repository
public interface IShiftSwapRequestRepository {
    void save(ShiftSwapRequest request);
    Optional<ShiftSwapRequest> findById(SwapRequestId id);
    List<ShiftSwapRequest> findByRequesterId(String requesterId);
    List<ShiftSwapRequest> findByCounterpartId(String counterpartId);
}
```

7.1.7 ShiftSchedule聚合根 (排班表)

職責: 記錄員工在某日被指派的班別，支援手動排班與輪班模式自動排班。

欄位:

- `id: ScheduleId` — 排班表 ID
- `employeeId: String` — 員工 ID
- `shiftId: ShiftId` — 班別 ID
- `scheduleDate: LocalDate` — 排班日期
- `status: ScheduleStatus` — 排班狀態 (DRAFT/PUBLISHED/LOCKED)
- `rotationPatternId: String` — 輪班模式 ID（手動排班為 null）
- `note: String` — 備註

Domain 行為:

- `changeShift(ShiftId)` — 變更班別（LOCKED 狀態不可變更）
- `publish()` — 發佈排班（DRAFT → PUBLISHED）
- `lock()` — 鎖定排班（PUBLISHED → LOCKED）
- `revertToDraft()` — 退回草稿

7.1.8 RotationPattern聚合根 (輪班模式)

職責: 定義循環的班別序列（如：早班→中班→晚班→休息→早班...）

欄位:

- `id: RotationPatternId` — 輪班模式 ID
- `organizationId: String` — 組織 ID
- `name/code: String` — 名稱/代碼
- `cycleDays: int` — 循環天數
- `rotationDays: List<RotationDay>` — 循環天序（每天的班別或休息）

Domain 行為:

- `setRotationDays(List<RotationDay>)` — 整批設定天序
- `getDayForIndex(int dayIndex)` — 取得某日應排的班別

RotationDay 內部值物件:

- `dayOrder: int` — 循環中第幾天（1-based）
- `shiftId: ShiftId` — 班別（休息日為 null）
- `isRestDay: boolean` — 是否為休息日

7.1.9 ShiftSwapRequest聚合根 (換班申請)

職責: 管理員工間的換班流程（申請 → 對方同意 → 主管審核）

狀態流轉:

```
PENDING_COUNTERPART → acceptByCounterpart() → PENDING_APPROVAL
PENDING_COUNTERPART → rejectByCounterpart() → REJECTED
PENDING_APPROVAL → approve() → APPROVED
PENDING_APPROVAL → reject() → REJECTED
(任何未結案狀態) → cancel() → CANCELLED
```

7.1.10 ShiftSchedulingDomainService (排班 Domain Service)

職責: 根據輪班模式自動產生排班表、執行換班交換

方法:

- `generateSchedules(employeeId, pattern, startDate, endDate, rotationStart)` — 自動排班產生
- `executeSwap(scheduleA, scheduleB)` — 執行兩筆排班的班別交換

---

8. 領域事件設計

8.1 事件清單

事件名稱	觸發時機	發布服務	訂閱服務
`AttendanceRecorded`	員工打卡	Attendance	-
`AttendanceAnomalyDetected`	偵測到打卡異常	Attendance	Notification
`LeaveApplied`	請假申請提交	Attendance	Workflow
`LeaveApproved`	請假審核通過	Attendance	Payroll, Notification
`LeaveRejected`	請假審核駁回	Attendance	Notification
`LeaveCancelled`	請假取消	Attendance	-
`OvertimeApplied`	加班申請提交	Attendance	Workflow
`OvertimeApproved`	加班審核通過	Attendance	Payroll
`OvertimeLimitExceeded`	加班時數超過上限	Attendance	Notification
`AnnualLeaveExpiring`	特休即將到期	Attendance	Notification
`AttendanceMonthClosed`	月度差勤結算	Attendance	Payroll, Report
`CompTimeGrantedEvent`	加班核准選擇補休 [2026-03-17 新增]	Attendance	Attendance (self)
`CompTimeExpiredEvent`	補休到期自動折算 [2026-03-17 新增]	Attendance	Payroll
`CompTimeExpiringEvent`	補休即將到期 [2026-03-17 新增]	Attendance	Notification
`CompTimeSettlementEvent`	離職補休結算 [2026-03-17 新增]	Attendance	Payroll
`AnnualLeaveSettlementEvent`	離職特休結算 [2026-03-17 新增]	Attendance	Payroll

8.2 事件Schema與範例

8.2.1 AttendanceRecordedEvent

```
{
  "eventId": "evt-att-001",
  "eventType": "AttendanceRecorded",
  "timestamp": "2025-12-06T09:05:00Z",
  "payload": {
    "recordId": "550e8400-e29b-41d4-a716-446655440000",
    "employeeId": "emp-001",
    "recordDate": "2025-12-06",
    "checkInTime": "2025-12-06T09:05:00",
    "checkOutTime": null,
    "isLate": true,
    "lateMinutes": 5
  }
}
```

8.2.2 LeaveApprovedEvent

```
{
  "eventId": "evt-lv-002",
  "eventType": "LeaveApproved",
  "timestamp": "2025-12-06T10:30:00Z",
  "payload": {
    "applicationId": "app-leave-001",
    "employeeId": "emp-001",
    "leaveTypeId": "lt-annual",
    "leaveTypeName": "特休假",
    "startDate": "2025-12-09",
    "endDate": "2025-12-10",
    "totalDays": 2,
    "isPaid": true,
    "payRate": 1.0,
    "approverId": "mgr-001",
    "approvedAt": "2025-12-06T10:30:00"
  }
}
```

8.2.3 OvertimeApprovedEvent

```
{
  "eventId": "evt-ot-003",
  "eventType": "OvertimeApproved",
  "timestamp": "2025-12-06T11:00:00Z",
  "payload": {
    "overtimeId": "ot-001",
    "employeeId": "emp-001",
    "overtimeDate": "2025-12-05",
    "overtimeHours": 2.5,
    "overtimeType": "WEEKDAY",
    "compensationType": "PAY",
    "approverId": "mgr-001"
  }
}
```

8.2.4 AttendanceMonthClosedEvent (關鍵事件)

```
{
  "eventId": "evt-close-004",
  "eventType": "AttendanceMonthClosed",
  "timestamp": "2025-12-01T00:00:00Z",
  "payload": {
    "month": "2025-11",
    "employeeId": "emp-001",
    "summary": {
      "totalWorkDays": 22,
      "actualWorkDays": 20,
      "totalLeaveDays": 2,
      "totalOvertimeHours": 15.5,
      "lateCount": 3,
      "lateTotalMinutes": 25,
      "earlyLeaveCount": 1,
      "earlyLeaveTotalMinutes": 10
    },
    "leaveDetails": [
      {"leaveType": "ANNUAL", "days": 1.5, "isPaid": true},
      {"leaveType": "SICK", "days": 0.5, "isPaid": false}
    ],
    "overtimeDetails": [
      {"type": "WEEKDAY", "hours": 10.5, "compensation": "PAY"},
      {"type": "REST_DAY", "hours": 5, "compensation": "PAY"}
    ]
  }
}
```

9. API設計

9.1 API總覽

模組	API數量	說明
打卡管理	6	打卡、查詢記錄、補卡申請
請假管理	8	餘額、申請、審核、取消
加班管理	6	申請、審核、統計
班別管理	4	CRUD
假別管理	4	CRUD
報表結算	3	月報、結算
合計	**31**	

9.2 Controller命名對照 (符合命名規範)

Controller	說明
`HR03CheckInCmdController`	打卡Command操作
`HR03CheckInQryController`	打卡記錄Query操作
`HR03LeaveCmdController`	請假Command操作
`HR03LeaveQryController`	請假Query操作
`HR03OvertimeCmdController`	加班Command操作
`HR03OvertimeQryController`	加班Query操作
`HR03ShiftCmdController`	班別管理Command操作
`HR03ShiftQryController`	班別Query操作
`HR03ReportQryController`	報表Query操作
`HR03MonthCloseCmdController`	月結Command操作

9.3 打卡管理API

9.3.1 上班打卡

端點: `POST /api/v1/attendance/check-in`

Controller: `HR03CheckInCmdController`

Service: `CheckInServiceImpl`

Request:

```
{
  "employeeId": "550e8400-e29b-41d4-a716-446655440000",
  "checkInTime": "2025-12-06T09:05:00",
  "location": {
    "latitude": 25.0330,
    "longitude": 121.5654
  },
  "ipAddress": "192.168.1.100"
}
```

Response 200:

```
{
  "recordId": "550e8400-e29b-41d4-a716-446655440001",
  "checkInTime": "2025-12-06T09:05:00",
  "isLate": true,
  "lateMinutes": 5,
  "shift": {
    "shiftName": "標準班",
    "workStartTime": "09:00",
    "lateToleranceMinutes": 0
  }
}
```

錯誤碼:

HTTP狀態碼	錯誤碼	說明
400	ALREADY_CHECKED_IN	今日已完成上班打卡
400	LOCATION_OUT_OF_RANGE	位置不在允許範圍內
404	NO_SHIFT_ASSIGNED	員工未設定班別

9.3.2 下班打卡

端點: `POST /api/v1/attendance/check-out`

Service: `CheckOutServiceImpl`

Request:

```
{
  "employeeId": "550e8400-e29b-41d4-a716-446655440000",
  "checkOutTime": "2025-12-06T18:30:00",
  "location": {
    "latitude": 25.0330,
    "longitude": 121.5654
  },
  "ipAddress": "192.168.1.100"
}
```

Response 200:

```
{
  "recordId": "550e8400-e29b-41d4-a716-446655440001",
  "checkInTime": "2025-12-06T09:05:00",
  "checkOutTime": "2025-12-06T18:30:00",
  "workingHours": 8.42,
  "isEarlyLeave": false
}
```

9.4 請假管理API

9.4.1 查詢假期餘額

端點: `GET /api/v1/leave/balances?employeeId={id}`

Controller: `HR03LeaveQryController`

Service: `GetLeaveBalancesServiceImpl`

Response 200:

```json
{
  "employeeId": "emp-001",
  "balances": [
    {
      "leaveTypeId": "lt-annual",
      "leaveTypeName": "特休假",
      "year": 2025,
      "totalDays": 10,
      "usedDays": 3.5,
      "remainingDays": 6.5,
      "expiryDate": "2026-12-31"
    },
    {
      "leaveTypeId": "lt-sick",
      "leaveTypeName": "病假",
      "year": 2025,
      "totalDays": 30,
      "usedDays": 2,
      "remainingDays": 28,
      "expiryDate": null
    }
  ]
}
```

9.4.2 提交請假申請

端點: `POST /api/v1/leave/applications`

Controller: `HR03LeaveCmdController`

Service: `CreateLeaveApplicationServiceImpl`

Request:

```json
{
  "employeeId": "emp-001",
  "leaveTypeId": "lt-annual",
  "startDate": "2025-12-09",
  "endDate": "2025-12-10",
  "startPeriod": "FULL_DAY",
  "endPeriod": "FULL_DAY",
  "reason": "家庭事務",
  "proofAttachmentUrl": null
}
```

Response 201:

```
{
  "applicationId": "app-001",
  "totalDays": 2,
  "remainingBalance": 4.5,
  "status": "PENDING",
  "workflowInstanceId": "wf-001"
}
```

錯誤碼:

HTTP狀態碼	錯誤碼	說明
400	INSUFFICIENT_BALANCE	假期餘額不足
400	DATE_OVERLAP	日期與其他申請重疊
400	PROOF_REQUIRED	需要上傳證明文件

9.4.3 審核請假 (核准)

端點: `PUT /api/v1/leave/applications/{id}/approve`

Controller: `HR03LeaveCmdController`

Service: `ApproveLeaveServiceImpl`

權限: `attendance:leave:approve`

Response 200:

```
{
  "applicationId": "app-001",
  "status": "APPROVED",
  "approvedBy": "李經理",
  "approvedAt": "2025-12-06T10:30:00Z"
}
```

9.5 加班管理API

9.5.1 提交加班申請

端點: `POST /api/v1/overtime/applications`

Controller: `HR03OvertimeCmdController`

Service: `CreateOvertimeApplicationServiceImpl`

Request:

```
{
  "employeeId": "emp-001",
  "overtimeDate": "2025-12-06",
  "startTime": "18:00",
  "endTime": "20:30",
  "overtimeType": "WEEKDAY",
  "reason": "專案趕工",
  "compensationType": "PAY"
}
```

Response 201:

```
{
  "overtimeId": "ot-001",
  "overtimeHours": 2.5,
  "status": "PENDING",
  "monthlyStatistics": {
    "accumulatedHours": 15.5,
    "monthlyLimit": 46,
    "quarterlyAccumulatedHours": 45.5,
    "quarterlyLimit": 138
  }
}
```

9.5.2 查詢加班統計

端點: `GET /api/v1/overtime/statistics?employeeId={id}&month={month}`

Controller: `HR03OvertimeQryController`

Service: `GetOvertimeStatisticsServiceImpl`

Response 200:

```
{
  "employeeId": "emp-001",
  "month": "2025-12",
  "totalHours": 15.5,
  "byType": {
    "WEEKDAY": 12.0,
    "REST_DAY": 3.5,
    "HOLIDAY": 0
  },
  "monthlyLimit": 46,
  "quarterlyAccumulatedHours": 45.5,
  "quarterlyLimit": 138,
  "warnings": []
}
```

9.6 月度結算API

9.6.1 執行月度結算

端點: `POST /api/v1/attendance/monthly-close`

Controller: `HR03MonthCloseCmdController`

Service: `MonthlyCloseServiceImpl`

權限: `attendance:close`

Request:

```
{
  "month": "2025-11",
  "organizationId": "org-001"
}
```

Response 200:

```
{
  "month": "2025-11",
  "processedEmployees": 150,
  "closedAt": "2025-12-01T00:00:00Z"
}
```

後續事件:

- ✅ 為每位員工發布 `AttendanceMonthClosed` 事件
- ✅ Payroll Service訂閱事件進行薪資計算

9.8 曠職自動判定排程 (AbsentDetectionJob)

排程: 每日 19:00（ `cron: 0 0 19 * * ?` ）

流程:

1. 取得所有在職員工 ID（employee_read_models / leave_balances 降級）
2. 查詢今日已有打卡記錄的員工 ID
3. 查詢今日有核准請假的員工 ID
4. AbsentDetectionDomainService 比對差集 → 缺勤員工清單
5. 為每位缺勤員工建立 AttendanceRecord (anomalyType=ABSENT)
6. 發布 `AttendanceAnomalyDetectedEvent` → 通知 HR/主管

領域服務: `AbsentDetectionDomainService`

- `detectAbsentEmployees(allIds, withRecord, onLeave)` → 缺勤員工 ID 清單
- `createAbsentRecord(employeeId, date)` → AttendanceRecord (ABSENT)

10. 工項清單摘要

前端開發工項

1. HR03-P01 打卡頁面 (GPS定位、即時時間)
2. HR03-P02 差勤記錄頁面 (行事曆、統計)
3. HR03-P03 請假申請頁面 (餘額顯示、表單)
4. HR03-P04 假期餘額頁面
5. HR03-P05 加班申請頁面 (時數預警)
6. HR03-P06 差勤審核頁面 (主管)
7. HR03-P07 班別管理頁面
8. HR03-P08 假別管理頁面
9. HR03-P09 差勤報表頁面
10. HR03-P10 月度結算頁面

後端開發工項

1. 班別聚合根與Repository
2. 打卡記錄聚合根與Repository
3. 請假申請聚合根與Repository
4. 加班申請聚合根與Repository
5. 假期餘額聚合根與Repository
6. 特休假政策聚合根
7. 打卡管理API (6端點)
8. 請假管理API (8端點)
9. 加班管理API (6端點)
10. 班別/假別管理API (8端點)
11. 報表結算API (3端點)
12. Workflow整合 (請假/加班審核)
13. 排程任務 (特休到期提醒、月結算Job)
14. 曠職自動判定排程 (AbsentDetectionJob) + 通知

資料庫開發工項

1. 建立8個資料表DDL
2. 建立索引
3. 初始化法定假別與預設班別
4. 初始化特休假規則 (依勞基法)

11. 特休/補休法規合規性設計 [2026-03-17 更新]

11.1 AnnualLeaveRule 非整數年資支援

問題： 現有 `AnnualLeaveRule` 的 `minServiceYears` / `maxServiceYears` 雖已宣告為 `DECIMAL(4,2)`，但 Domain 計算邏輯中以 `int` 處理年資，無法正確處理 6 個月（0.5 年）的情境。

設計變更：

11.1.1 AnnualLeaveRule 欄位調整

保留現有 `min_service_years DECIMAL(4,2)` / `max_service_years DECIMAL(4,2)` 資料庫欄位不變，但 Domain 層計算需改用 `BigDecimal` 精確比對：

```java
/**
 * 特休規則 — 支援非整數年資（如 0.5 年 = 6 個月）
 * [2026-03-17 更新] 修正年資比對邏輯以符合勞基法第 38 條
 */
@Entity
@Table(name = "annual_leave_rules")
public class AnnualLeaveRule {
    @EmbeddedId
    private RuleId id;

    @Column(name = "policy_id", nullable = false)
    private UUID policyId;

    // 已為 BigDecimal，無需修改資料庫
    @Column(name = "min_service_years", nullable = false, precision = 4, scale = 2)
    private BigDecimal minServiceYears;

    @Column(name = "max_service_years", precision = 4, scale = 2)
    private BigDecimal maxServiceYears; // null 表示無上限（10 年以上）

    @Column(name = "leave_days", nullable = false)
    private int leaveDays;

    /**
     * 檢查年資是否落在此規則區間
     * 區間為 [minServiceYears, maxServiceYears)，左閉右開
     */
    public boolean matches(BigDecimal serviceYears) {
        boolean aboveMin = serviceYears.compareTo(minServiceYears) >= 0;
        boolean belowMax = maxServiceYears == null
            || serviceYears.compareTo(maxServiceYears) < 0;
        return aboveMin && belowMax;
    }
}
```

11.1.2 AnnualLeavePolicy 計算邏輯修正

```java
/**
 * 計算特休天數（依勞基法第 38 條）
 * [2026-03-17 更新] 修正為精確年資比對，支援非整數年資
 */
public int calculateAnnualLeaveDays(LocalDate hireDate, LocalDate currentDate) {
    BigDecimal serviceYears = calculateServiceYears(hireDate, currentDate);

    // 未滿 6 個月，無特休
    if (serviceYears.compareTo(new BigDecimal("0.5")) < 0) {
        return 0;
    }

    // 依規則表逐一比對
    for (AnnualLeaveRule rule : rules) {
        if (rule.matches(serviceYears)) {
            // 10 年以上特殊邏輯：base 15 天 + 每年加 1 天，上限 30 天
            if (serviceYears.compareTo(BigDecimal.TEN) >= 0) {
                int extraYears = serviceYears.intValue() - 10;
                return Math.min(rule.getLeaveDays() + extraYears, 30);
            }
            return rule.getLeaveDays();
        }
    }

    return 0;
}

/**
 * 精確計算年資（以月為基礎，避免整數截斷）
 * [2026-03-17 更新]
 */
private BigDecimal calculateServiceYears(LocalDate hireDate, LocalDate currentDate) {
    long totalMonths = ChronoUnit.MONTHS.between(hireDate, currentDate);
    return new BigDecimal(totalMonths)
        .divide(new BigDecimal("12"), 2, RoundingMode.DOWN);
}
```

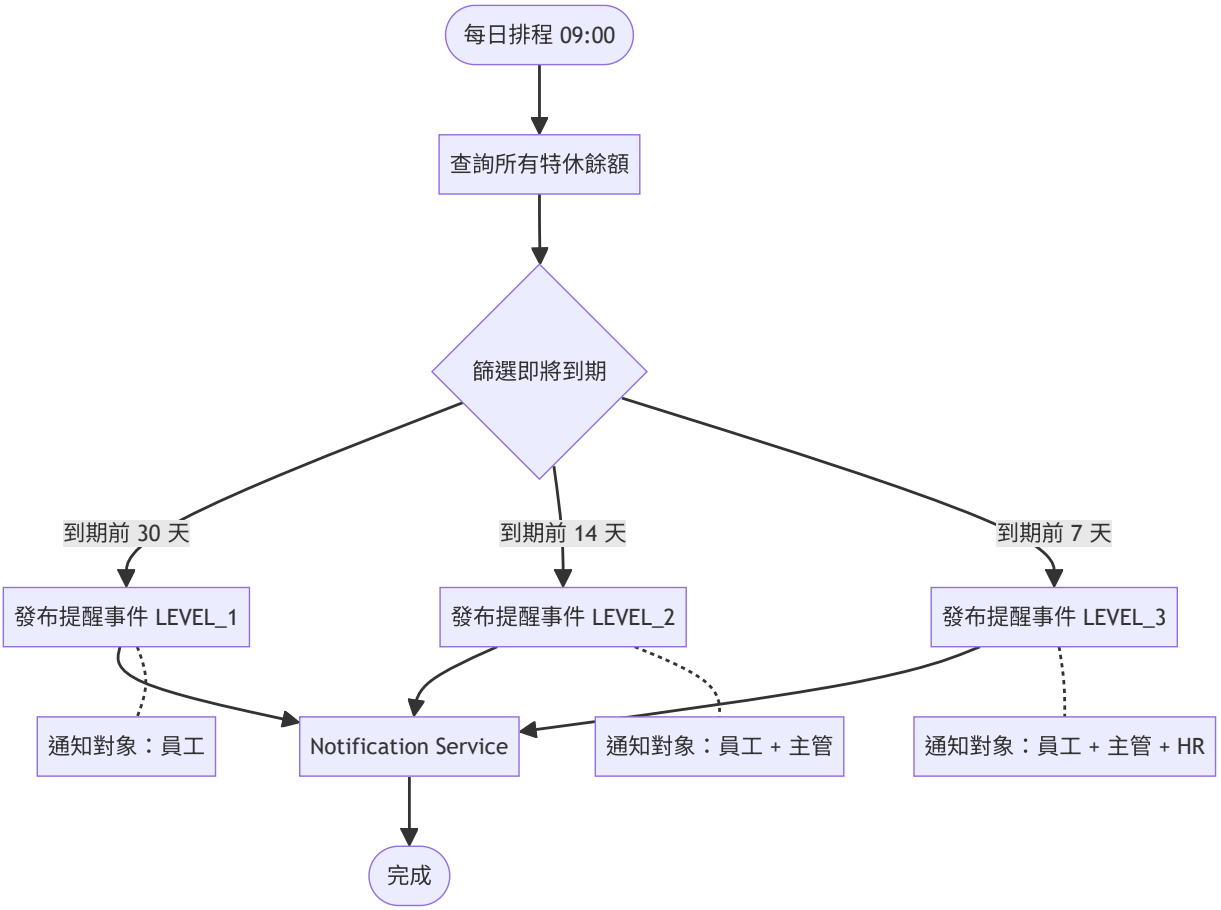
11.1.3 預設規則初始資料修正

```sql
-- [2026-03-17 更新] 確認 min_service_years 使用精確小數
-- 現有 SQL 已使用 0.5，無需變更，但需確認 10 年以上的 leave_days = 15（base）
-- 10 年以上每年 +1 天由程式邏輯處理，資料庫只記錄 base 值
UPDATE annual_leave_rules
SET leave_days = 15
WHERE min_service_years = 10.00
  AND policy_id = (SELECT policy_id FROM annual_leave_policies LIMIT 1);
```

11.2 特休使用提醒機制設計

法規依據： 勞基法第 38 條第 3 項 — 雇主應主動提醒勞工排定特休。

11.2.1 排程設計 — AnnualLeaveReminderJob



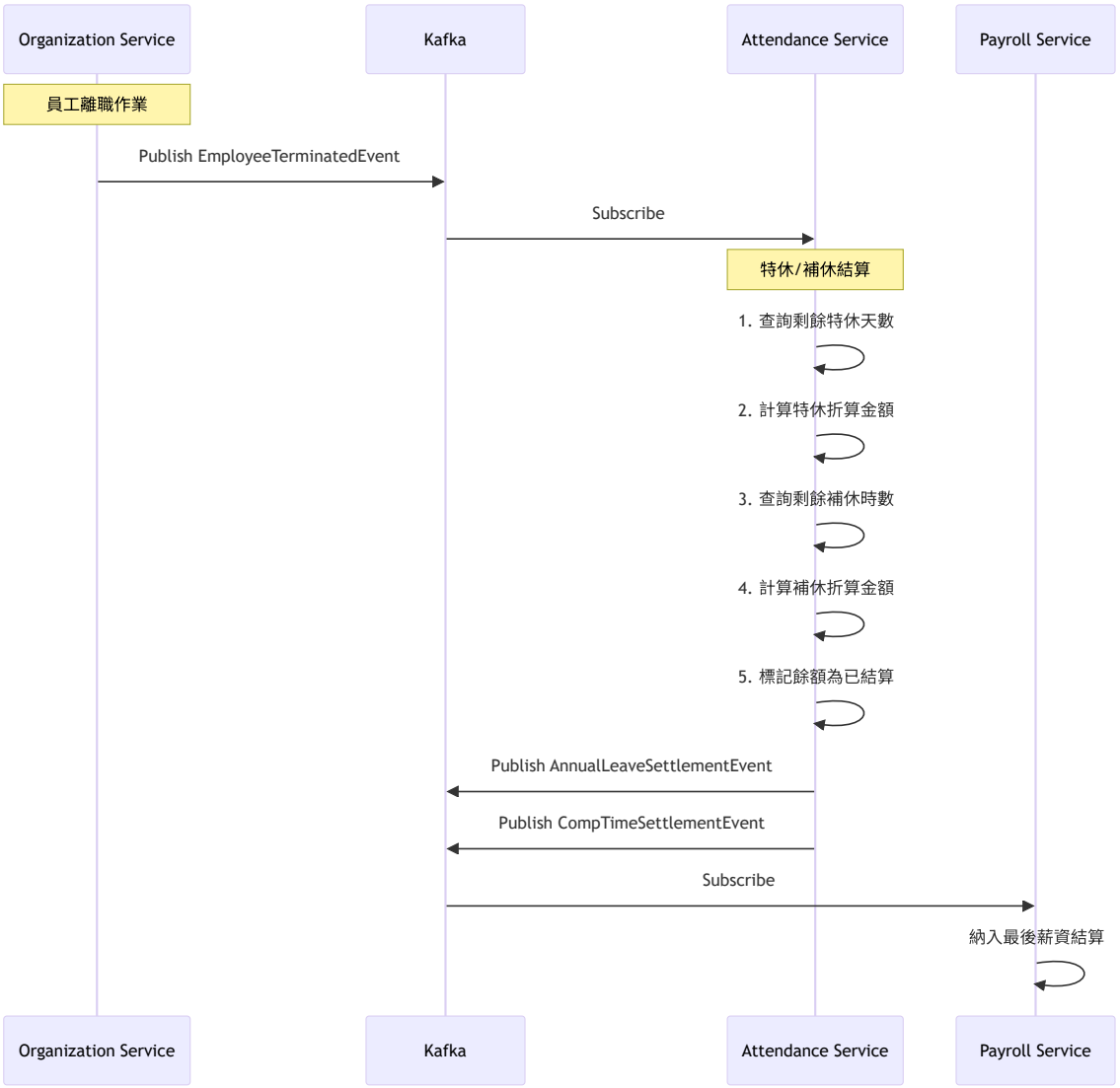
11.2.2 提醒事件 Schema

```
{
  "eventType": "AnnualLeaveExpiring",
  "payload": {
    "employeeId": "uuid",
    "employeeName": "張三",
    "remainingDays": 5.0,
    "expiryDate": "2026-12-31",
    "daysUntilExpiry": 14,
    "reminderLevel": "LEVEL_2",
    "notifyTargets": ["EMPLOYEE", "MANAGER"]
  }
}
```

11.3 離職特休結算設計

法規依據： 勞基法第 38 條第 4 項

11.3.1 事件驅動架構



11.3.2 結算 Domain Service

```java
/**
 * 離職特休/補休結算 Domain Service
 * [2026-03-17 新增]
 */
public class TerminationLeaveSettlementDomainService {

    /**
     * 計算特休折算金額
     * @param remainingDays 剩餘特休天數
     * @param dailyWage 日薪（月薪 / 30）
     * @return 折算金額
     */
    public BigDecimal calculateAnnualLeaveSettlement(
            BigDecimal remainingDays, BigDecimal dailyWage) {
        if (remainingDays.compareTo(BigDecimal.ZERO) <= 0) {
            return BigDecimal.ZERO;
        }
        return remainingDays.multiply(dailyWage)
            .setScale(0, RoundingMode.CEILING); // 無條件進位至整數
    }

    /**
     * 計算補休折算金額
     * @param compTimeBalances 所有未結算的補休餘額
     * @return 折算總金額
     */
    public BigDecimal calculateCompTimeSettlement(
            List<CompTimeBalance> compTimeBalances) {
        return compTimeBalances.stream()
            .filter(b -> b.getRemainingHours().compareTo(BigDecimal.ZERO) > 0)
            .map(CompTimeBalance::calculateSettlementAmount)
            .reduce(BigDecimal.ZERO, BigDecimal::add)
            .setScale(0, RoundingMode.CEILING);
    }
}
```

11.3.3 Application Service Pipeline

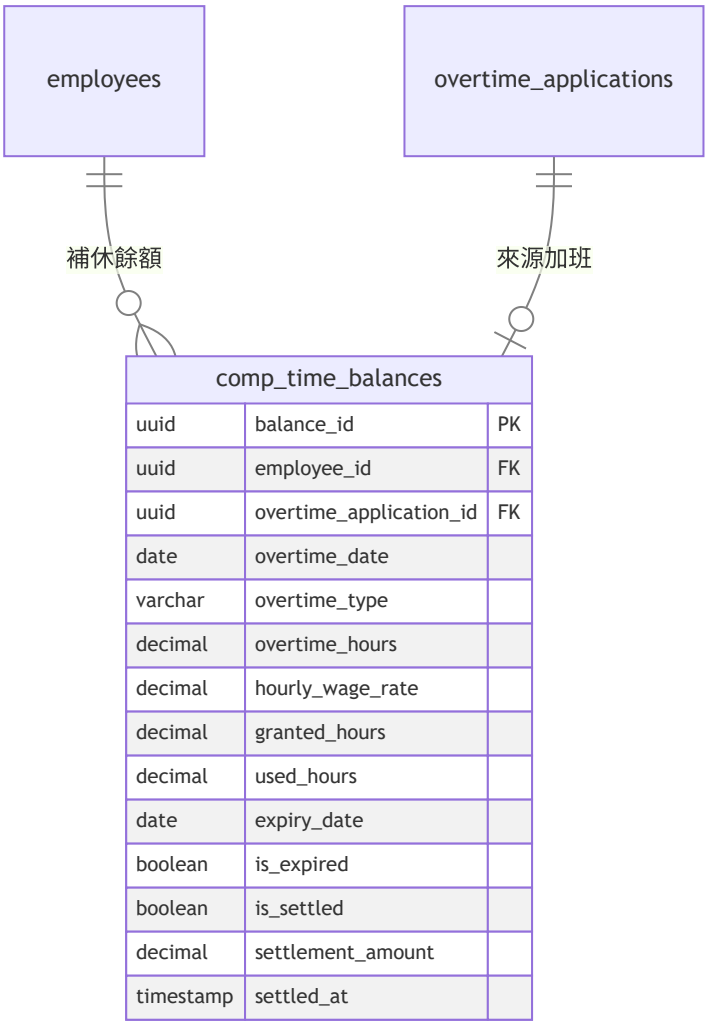
```
/**
 * 離職結算 Service — 監聽 EmployeeTerminatedEvent
 * [2026-03-17 新增]
 *
 * Pipeline Tasks:
 * 1. LoadTerminationDataTask  — 載入員工特休/補休餘額
 * 2. CalculateSettlementTask  — 計算折算金額
 * 3. MarkBalancesSettledTask  — 標記餘額為已結算
 * 4. PublishSettlementEventsTask — 發布結算事件給 Payroll
 */
@Service("terminationLeaveSettlementServiceImpl")
public class TerminationLeaveSettlementServiceImpl {
    // 由 EmployeeTerminatedEvent 的 Kafka Consumer 觸發
}
```

11.4 補休餘額（CompTimeBalance）設計

11.4.1 資料庫 Schema

```
-- [2026-03-17 新增] 補休餘額表
CREATE TABLE comp_time_balances (
    balance_id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    employee_id UUID NOT NULL,
    overtime_application_id UUID NOT NULL REFERENCES overtime_applications(overtime_id),

    -- 來源加班資訊
    overtime_date DATE NOT NULL,
    overtime_type VARCHAR(20) NOT NULL
        CHECK (overtime_type IN ('WEEKDAY', 'REST_DAY', 'HOLIDAY')),
    overtime_hours DECIMAL(4,1) NOT NULL,
    hourly_wage_rate DECIMAL(10,2) NOT NULL,

    -- 補休額度
    granted_hours DECIMAL(4,1) NOT NULL,
    used_hours DECIMAL(4,1) NOT NULL DEFAULT 0,

    -- 到期管理
    expiry_date DATE NOT NULL,
    is_expired BOOLEAN DEFAULT FALSE,
    is_settled BOOLEAN DEFAULT FALSE,
    settlement_amount DECIMAL(10,0),
    settled_at TIMESTAMP,

    created_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP DEFAULT CURRENT_TIMESTAMP,

    CONSTRAINT chk_comp_hours CHECK (used_hours <= granted_hours),
    CONSTRAINT chk_comp_ot_type CHECK (overtime_type IN ('WEEKDAY', 'REST_DAY', 'HOLIDAY'))
);

CREATE INDEX idx_comp_time_employee ON comp_time_balances(employee_id);
CREATE INDEX idx_comp_time_expiry ON comp_time_balances(expiry_date)
    WHERE is_expired = FALSE AND is_settled = FALSE;
CREATE INDEX idx_comp_time_overtime ON comp_time_balances(overtime_application_id);

COMMENT ON TABLE comp_time_balances IS '補休餘額表 [2026-03-17 新增]';
COMMENT ON COLUMN comp_time_balances.hourly_wage_rate IS '加班當日的時薪費率，供到期折算使用';
COMMENT ON COLUMN comp_time_balances.expiry_date IS '補休到期日，依勞基法第32-1條不得超過年度結
COMMENT ON COLUMN comp_time_balances.settlement_amount IS '到期/離職折算金額';
```

11.4.2 ER Diagram 補充



11.4.3 CompTimeBalance Domain 設計

```java
/**
 * 補休餘額聚合根
 * [2026-03-17 新增]
 * 法規依據：勞基法第 32-1 條
 */
@Entity
@Table(name = "comp_time_balances")
public class CompTimeBalance {
    @EmbeddedId
    private CompTimeBalanceId id;

    @Column(name = "employee_id", nullable = false)
    private UUID employeeId;

    @Column(name = "overtime_application_id", nullable = false)
    private UUID overtimeApplicationId;

    @Column(name = "overtime_date", nullable = false)
    private LocalDate overtimeDate;

    @Enumerated(EnumType.STRING)
    @Column(name = "overtime_type", nullable = false)
    private OvertimeType overtimeType;

    @Column(name = "overtime_hours", nullable = false)
    private BigDecimal overtimeHours;

    @Column(name = "hourly_wage_rate", nullable = false)
    private BigDecimal hourlyWageRate;

    @Column(name = "granted_hours", nullable = false)
    private BigDecimal grantedHours;

    @Column(name = "used_hours", nullable = false)
    private BigDecimal usedHours;

    @Column(name = "expiry_date", nullable = false)
    private LocalDate expiryDate;

    @Column(name = "is_expired")
    private boolean isExpired;

    @Column(name = "is_settled")
    private boolean isSettled;

    @Column(name = "settlement_amount")
    private BigDecimal settlementAmount;

    @Column(name = "settled_at")
```

```java
    private LocalDateTime settledAt;

    // ========== 計算屬性 ==========

    /**
     * 剩餘可用補休時數
     */
    public BigDecimal getRemainingHours() {
        return grantedHours.subtract(usedHours);
    }

    // ========== Domain 行為 ==========

    /**
     * 建立補休餘額（加班核准且選擇補休時呼叫）
     */
    public static CompTimeBalance create(
            UUID employeeId,
            OvertimeApplication overtime,
            BigDecimal hourlyWageRate,
            LocalDate expiryDate) {

        CompTimeBalance balance = new CompTimeBalance();
        balance.id = CompTimeBalanceId.generate();
        balance.employeeId = employeeId;
        balance.overtimeApplicationId = overtime.getId().getValue();
        balance.overtimeDate = overtime.getOvertimeDate();
        balance.overtimeType = overtime.getOvertimeType();
        balance.overtimeHours = overtime.getOvertimeHours();
        balance.hourlyWageRate = hourlyWageRate;
        balance.grantedHours = overtime.getOvertimeHours(); // 1:1 換算
        balance.usedHours = BigDecimal.ZERO;
        balance.expiryDate = expiryDate;
        balance.isExpired = false;
        balance.isSettled = false;

        return balance;
    }

    /**
     * 使用補休時數
     */
    public void use(BigDecimal hours) {
        if (this.isExpired || this.isSettled) {
            throw new DomainException("此補休已到期或已結算，無法使用");
        }
        BigDecimal remaining = getRemainingHours();
        if (remaining.compareTo(hours) < 0) {
            throw new DomainException(
                String.format("補休時數不足，剩餘 %s 小時，申請 %s 小時",
                    remaining, hours));
```

```java
        }
        this.usedHours = this.usedHours.add(hours);
    }

    /**
     * 到期處理 — 未休時數折算為加班費
     * 折算金額 = 剩餘時數 x 時薪費率 x 加班費率倍數
     */
    public BigDecimal expire() {
        if (this.isSettled) {
            throw new DomainException("此補休已結算");
        }

        this.isExpired = true;
        this.isSettled = true;
        this.settledAt = LocalDateTime.now();

        BigDecimal remaining = getRemainingHours();
        if (remaining.compareTo(BigDecimal.ZERO) <= 0) {
            this.settlementAmount = BigDecimal.ZERO;
            return BigDecimal.ZERO;
        }

        this.settlementAmount = calculateSettlementAmount();
        return this.settlementAmount;
    }

    /**
     * 離職結算 — 邏輯同到期處理
     */
    public BigDecimal settleOnTermination() {
        return expire();
    }

    /**
     * 計算折算金額
     * 依加班當日工資費率折算（勞基法第 32-1 條）
     */
    public BigDecimal calculateSettlementAmount() {
        BigDecimal remaining = getRemainingHours();
        BigDecimal overtimeMultiplier = getOvertimeMultiplier(this.overtimeType);
        return remaining.multiply(hourlyWageRate)
            .multiply(overtimeMultiplier)
            .setScale(0, RoundingMode.CEILING);
    }

    /**
     * 取得加班費率倍數
     * 平日延長工時前 2 小時: 1.34, 後 2 小時: 1.67
     * 休息日: 前 2 小時 1.34, 2~8 小時 1.67, 8 小時以上 2.67
     * 國定假日: 2.0
```

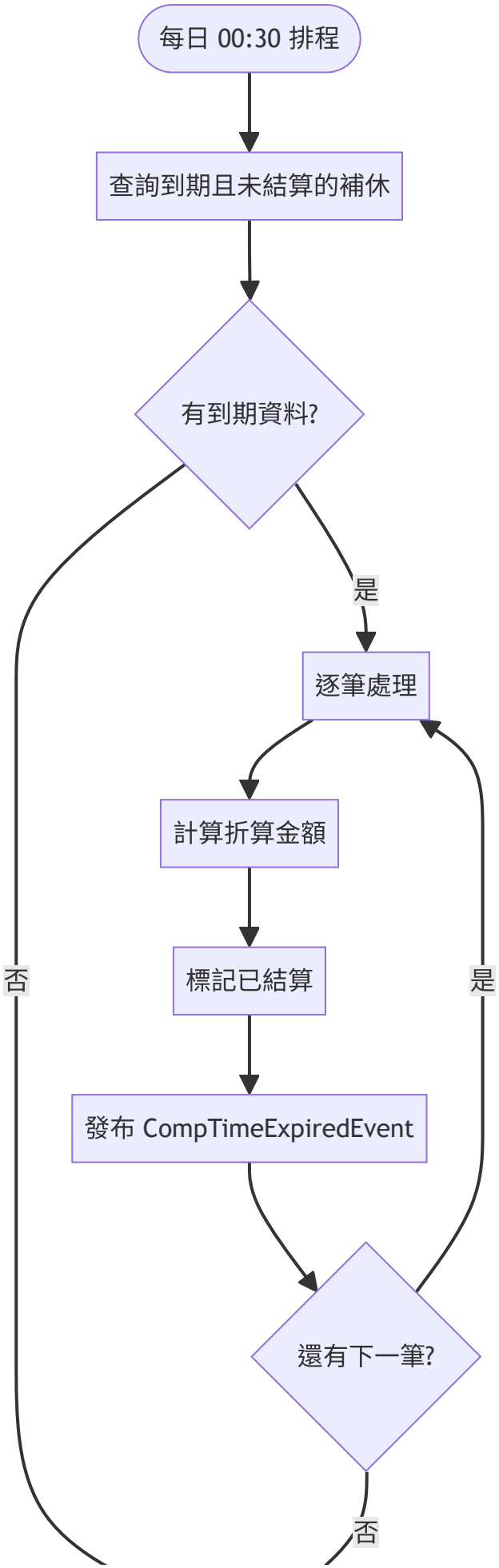
```java
     * 簡化處理：使用加權平均倍數
     */
    private BigDecimal getOvertimeMultiplier(OvertimeType type) {
        switch (type) {
            case WEEKDAY:  return new BigDecimal("1.34");
            case REST_DAY: return new BigDecimal("1.34");
            case HOLIDAY:  return new BigDecimal("2.00");
            default:       return new BigDecimal("1.34");
        }
    }

    /**
     * 檢查是否即將到期
     */
    public boolean isExpiringSoon(int daysThreshold) {
        if (this.isExpired || this.isSettled) return false;
        if (getRemainingHours().compareTo(BigDecimal.ZERO) <= 0) return false;

        LocalDate threshold = LocalDate.now().plusDays(daysThreshold);
        return this.expiryDate.isBefore(threshold)
            || this.expiryDate.isEqual(threshold);
    }
}
```

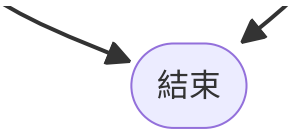
11.4.4 CompTimeBalance Repository

```java
/**
 * 補休餘額 Repository 介面
 * [2026-03-17 新增]
 */
public interface ICompTimeBalanceRepository {
    CompTimeBalance findById(CompTimeBalanceId id);
    List<CompTimeBalance> findByEmployeeId(UUID employeeId);
    List<CompTimeBalance> findActiveByEmployeeId(UUID employeeId);
    List<CompTimeBalance> findExpiredUnsettled(LocalDate asOfDate);
    List<CompTimeBalance> findExpiringSoon(LocalDate beforeDate);
    void save(CompTimeBalance balance);
}
```

11.5 排程任務設計

11.5.1 CompTimeExpiryJob（補休到期自動折算）


每日 00:30 排程
查詢到期且未結算的補休
有到期資料?
是
否
逐筆處理
計算折算金額
標記已結算
發布 CompTimeExpiredEvent
還有下一筆?
是
否



排程設定: `cron: 0 30 0 * * ?`

11.5.2 CompTimeReminderJob（補休到期提醒）

排程設定: `cron: 0 0 9 * * ?`

流程：

1. 查詢 7 天內即將到期、剩餘時數 > 0 的補休
2. 發布 `CompTimeExpiringEvent`
3. Notification Service 發送提醒

11.6 OvertimeApplication 聯動修改

加班申請核准且選擇補休時，需自動建立 CompTimeBalance：

```java
/**
 * 加班核准後處理
 * [2026-03-17 更新] 補充補休額度建立邏輯
 */
public void approve(UUID approverId) {
    if (this.status != ApplicationStatus.PENDING) {
        throw new DomainException("只能核准待審核的申請");
    }

    this.status = ApplicationStatus.APPROVED;

    // 若選擇補休，發布事件通知建立 CompTimeBalance
    if (this.compensationType == CompensationType.COMP_TIME) {
        DomainEventPublisher.publish(new CompTimeGrantedEvent(
            this.id.getValue(),
            this.employeeId,
            this.overtimeDate,
            this.overtimeHours,
            this.overtimeType
        ));
    } else {
        DomainEventPublisher.publish(new OvertimeApprovedEvent(
            this.id.getValue(),
            this.employeeId,
            this.overtimeHours,
            this.overtimeType,
            this.compensationType
        ));
    }
}
```

11.7 新增領域事件清單

事件名稱	觸發時機	發布服務	訂閱服務
`CompTimeGrantedEvent`	加班核准選擇補休	Attendance	Attendance (self)
`CompTimeExpiredEvent`	補休到期自動折算	Attendance	Payroll
`CompTimeExpiringEvent`	補休即將到期	Attendance	Notification
`CompTimeSettlementEvent`	離職補休結算	Attendance	Payroll
`AnnualLeaveSettlementEvent`	離職特休結算	Attendance	Payroll

11.8 新增事件 Schema

11.8.1 AnnualLeaveSettlementEvent

```
{
  "eventType": "AnnualLeaveSettlement",
  "timestamp": "2026-03-15T10:00:00Z",
  "payload": {
    "employeeId": "uuid-emp",
    "terminationDate": "2026-03-31",
    "remainingDays": 5.0,
    "dailyWage": 1667,
    "settlementAmount": 8335,
    "leaveYear": 2026
  }
}
```

11.8.2 CompTimeExpiredEvent

```
{
  "eventType": "CompTimeExpired",
  "timestamp": "2026-03-17T00:30:00Z",
  "payload": {
    "balanceId": "uuid-balance",
    "employeeId": "uuid-emp",
    "overtimeDate": "2026-01-15",
    "overtimeType": "WEEKDAY",
    "expiredHours": 3.0,
    "hourlyWageRate": 208,
    "overtimeMultiplier": 1.34,
    "settlementAmount": 837
  }
}
```

11.8.3 CompTimeSettlementEvent

```
{
  "eventType": "CompTimeSettlement",
  "timestamp": "2026-03-15T10:00:00Z",
  "payload": {
    "employeeId": "uuid-emp",
    "terminationDate": "2026-03-31",
    "settlements": [
      {
        "balanceId": "uuid-1",
        "overtimeDate": "2026-01-15",
        "overtimeType": "WEEKDAY",
        "expiredHours": 3.0,
        "amount": 837
      },
      {
        "balanceId": "uuid-2",
        "overtimeDate": "2026-02-20",
        "overtimeType": "REST_DAY",
        "expiredHours": 8.0,
        "amount": 2230
      }
    ],
    "totalSettlementAmount": 3067
  }
}
```

11.9 工項清單補充

後端開發工項（新增）

15. CompTimeBalance 聚合根與 Repository
16. TerminationLeaveSettlementDomainService（離職結算 Domain Service）
17. TerminationLeaveSettlementServiceImpl（監聽 EmployeeTerminatedEvent）
18. CompTimeExpiryJob 排程（每日檢查到期補休）
19. CompTimeReminderJob 排程（補休到期提醒）
20. AnnualLeaveReminderJob 排程修正（多級提醒 30/14/7 天）
21. OvertimeApplication 核准後補休建立邏輯
22. AnnualLeavePolicy.calculateAnnualLeaveDays() 修正非整數年資

資料庫開發工項（新增）

5. 建立 `comp_time_balances` 資料表 DDL
6. 建立相關索引
7. 確認 `annual_leave_rules` 的 10 年以上規則 base 值為 15

文件完成日期: 2025-12-06 **版本:** 1.1（2026-03-17 補充特休/補休法規合規性設計）